Supplementary Material

Architecture and Preprocessing Choices under Dataset Shift in Single-Channel EEG Sleep Staging: A Controlled Evaluation on Sleep-EDF and SHHS1

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This supplement provides model specifications, training settings and supporting statistical analyses. Unless stated otherwise, SHHS1 results refer to the first SHHS1 sample (180 participants). On that sample, seed 42 is primary and seed 123 is a post-protocol sensitivity repeat; E1 and E2 analyses are secondary, and E3−E2 and E4 analyses are post-hoc. Seed-specific results are kept separate and do not alter the primary comparison family. Supplementary Section S3 reports the confirmatory evaluation on the 500-participant holdout SHHS1 sample under a pre-specified protocol. Supplementary Section S4 examines approximate input-scale invariance and checkpoint similarities of E3 and E4. Supplementary Section S5 reports the post-confirmatory amplitude and AdaBN analyses, Supplementary Section S6 the second amendment (normalisation scope, five training seeds, single models, EM sample size and training cost), and Supplementary Section S7 a post-hoc description of the input power spectra, Supplementary Section S8 a stage-wise description of input amplitude, and Supplementary Section S9 the whole-night CPU timing. E6 estimates mean and standard deviation from each complete unlabelled target recording; it is label-free but transductive at recording level, not purely inductive zero-shot inference.

Supplementary Methods S1: model and training details

Architecture and feature construction

Each encoder input is a 30-s epoch with 3,000 samples. The control encoder contains 15 separately trained CNNs: three context groups (current, previous and next; C/P/N) crossed with five class-specialised loss functions. Every CNN has a five-class output, not a binary head. Concatenating its softmax outputs in C/P/N order gives 75 features per central epoch. The P and N inputs are shifted within each recording and are trained against the central epoch's label; the first or last signal epoch is replicated at a recording edge. No context crosses recording boundaries. These are softmax-output features; their use does not imply that probability calibration has been established.

The ResNet-1D encoder instead produces a 128-dimensional, pre-dropout representation of the current epoch. Layer specifications are given in Table S1. Encoders are trained and selected before sequence training. Their selected checkpoints are loaded in evaluation mode, and features are cached without gradients. E1 reuses the E0 encoder and its features within the same fold and seed; its sequence training settings differ from E0 (Table S2). E1−E0 is therefore a sequence-model configuration comparison, not an architecture-only effect. E2−E1 replaces the feature/context package, including encoder training and selection, while retaining the specified TCN recipe.

Table S1: Layer specifications. Convolution entries give input/output channels, kernel, stride and padding; d denotes dilation. All final classifiers have five outputs.

Component	Layers and operations
One control CNN	Conv1d $1 \to 10$, $k = 55$, $s = 1$, $p = 27$, no bias; ReLU; batch normalisation; max-pool $k = s = 16$, ceiling mode. Then Conv1d $10 \to 5$, $k = 25$, $s = 1$, $p = 12$, no bias; ReLU; batch normalisation; the same max-pool. Flatten $5 \times 12 = 60$; linear $60 \to 10$; ReLU; linear $10 \to 5$.
ResNet stem	Conv1d $1 \to 32$, $k = 50$, $s = 5$, $p = 25$, no bias; batch normalisation; ReLU; max-pool $k = 3$, $s = 2$, $p = 1$.
ResNet blocks	Three residual blocks: $32 \to 64$ ($s = 1$), $64 \to 128$ ($s = 2$), $128 \to 128$ ($s = 2$). Each has two bias-free $k = 7$, $p = 3$ convolutions, with the stated stride in the first and stride 1 in the second; batch normalisation after each, ReLU after the first and after residual addition. Each shortcut uses a bias-free $k = 1$ projection with the first convolution's stride, followed by batch normalisation.
ResNet output	Global average pooling gives 128 features. During encoder training only, dropout 0.5 precedes the linear $128 \to 5$ classifier. Cached features are taken before dropout.
BiLSTM	One bidirectional LSTM layer, input 75, hidden size 128 per direction; linear $256 \to 5$ output. Packed sequences exclude batch padding from recurrence.
TCN	Linear input projection 75 or $128 \to 128$; layer normalisation; GELU. Six residual blocks, each with two bias-free $128 \to 128$ convolutions, $k = 3$, $s = 1$, $p = d$, with $d \in \{1, 2, 4, 8, 16, 32\}$. Each convolution is followed by per-epoch layer normalisation, GELU and dropout 0.2. The shortcut is identity. Final dropout 0.2 and linear $128 \to 5$.

Training and checkpoint selection

All stages use the same subject partitions, with validation at the end of each epoch, but not the same optimisation or selection rule. For a CNN specialised to class c, training counts n_c and $N = \sum_c n_c$ determine weighted five-class cross-entropy: the weight for c is $0.5/(n_c/N)$ and each other class has weight $0.5/((N - n_c)/N)$. These training-derived weights are also used to evaluate its validation loss. ResNet uses weights $N/(5n_c)$, again computed only from training subjects. Sequence-model cross-entropy is unweighted over valid labelled epochs. CNN branch seeds are the campaign seed plus the branch index in C/P/N-stage order; ResNet and sequence training use the campaign seed. No label smoothing, data augmentation or learning-rate scheduler is used.

Table S2: Specified training recipes. Patience counts end-of-epoch validation checks without an improvement. E1 reuses the E0 encoder; all other reported encoder/sequence pairs are fitted for their own input variant.

Setting	CNN15 (E0/E1)	BiLSTM (E0)	ResNet (E2–E6)	TCN (E1–E6)
Optimiser	Adam	Adam	AdamW	Adam
Learning rate	10^{-3}	10^{-2}	10^{-3}	5×10^{-4}
Weight decay	0	0	10^{-4}	0
Batch size	64 epochs	4 recordings	64 epochs	8 recordings
Maximum epochs	1,000	1,000	40	300
Patience	10	10	8	30
Selected checkpoint	Minimum weighted validation loss	Maximum validation macro-F1	Maximum validation macro-F1	Maximum validation macro-F1
Gradient-norm clipping	None	1.0	None	1.0

Relation of E0 to the original ZleepAnlystNet

E0 follows the published ZleepAnlystNet design (Jirakittayakorn et al., Scientific Reports 14:9859, 2024): the 15 C/P/N class-specific CNNs, their layer sizes and the loss weights $0.5/p_i$, edge replication of the shifted P and N inputs, the 128-unit BiLSTM over 75 softmax features, and the optimiser, learning rates, batch sizes, patience and epoch limits of both stages. The original study was implemented in MATLAB without released code or seeds. The present reimplementation differs in five respects. (i) It uses PyTorch with bias-free convolutions and, for the BiLSTM, gradient-norm clipping at 1.0. (ii) Validation data are a held-out subject fold, whereas the original drew a random epoch-wise 70/30 split (CNNs) or 90/10 split (BiLSTM) from the training data. (iii) Validation runs at the end of every epoch instead of every 150 (CNN) or 10 (BiLSTM) iterations, and checkpoints are chosen by the rules in Table S2. (iv) Movement and unknown epochs are masked from the BiLSTM loss rather than relabelled as W. (v) Each rotation uses eight training folds, one validation fold and one test fold. The original headline results (accuracy 0.8702, macro-F1 0.8209) refer to Sleep-EDF-13 and are not comparable; its Sleep-EDF-18 Fpz-Cz results, obtained with ten-fold subject-wise cross-validation on nearly the same scored epochs, are compared with E0 in Table S3.

Table S3: E0 (seed 42, pooled out-of-fold) versus the published ZleepAnlystNet results on Sleep-EDF-18 Fpz-Cz (Fig. 3C of the original article). Values are proportions; differences are E0 minus published.

	Epochs	macro-F1	Accuracy	κ	F1 W	F1 N1	F1 N2	F1 N3	F1 REM
Published	195,479	.7826	.8409	.7790	.9311	.4831	.8586	.8116	.8285
E0	195,469	.7754	.8262	.7614	.9236	.5009	.8476	.7921	.8129
Difference	−10	−.0072	−.0147	−.0176	−.0075	+.0178	−.0110	−.0195	−.0156

The original study also applied its Sleep-EDF-18 Fpz-Cz models, without retraining, to 200 SHHS recordings scored from the C3-A2 channel and reported macro-F1 0.5344, κ 0.4859 and N3 F1 0.4286. The best-performing fold, selected on these SHHS recordings and applied to all 6,441 SHHS1 recordings, reached macro-F1 0.5735 and N3 F1 0.4580. These values are not directly comparable with the C4-A1 ensembles evaluated here, but they show a similar N3 deficit.

Whole retained recordings are the sequence units for training and inference. Batch padding and unknown/movement labels are excluded from the loss and metrics; unknown epochs retain their positions in the signal sequence. The TCN masks padded positions after the input projection and each convolutional sub-block. Its theoretical receptive field is $1 + 2(3 - -1)(1 + 2 + 4 + 8 + 16 + 32) = 253$ feature epochs, spanning up to 126 epochs on each side, subject to recording boundaries. This is not a causal architecture. E1 also receives the explicit neighbouring signal epochs through its CNN features. The 100-epoch input used for timing below is a benchmark shape, not a training-window length.

Signal operations and evaluation windows

Signal values are read in physical microvolts. Sleep-EDF uses Fpz-Cz at 100 Hz. SHHS1 uses C4-A1 and is resampled continuously from 125 to 100 Hz with polyphase resampling (up/down = 4/5, Kaiser window $\beta = 5$, constant padding).

Table S4 specifies operations after this cohort-specific loading/resampling step. Filtering uses a fourth-order Butterworth design with 0.5–30 Hz cutoffs, implemented as second-order sections with forward-backward filtering. It operates on the continuous recording before epoch extraction and trimming.

Table S4: Input variants and fitted components. Operations are listed in execution order; all model weights are learned on Sleep-EDF only.

ID	Signal operations	Encoder and sequence configuration
E0	Raw microvolt signal	CNN15 C/P/N, then BiLSTM
E1	Raw microvolt signal	Reused E0 CNN15 C/P/N, then TCN
E2	Raw microvolt signal	Current-epoch ResNet, then TCN
E3	Band-pass; clip at $\pm 800\,\mu$V; divide by 100	Current-epoch ResNet, then TCN
E4	Band-pass	Current-epoch ResNet, then TCN
E5	Band-pass; clip at $\pm 800\,\mu$V	Input-identity audit; excluded from performance analysis
E6	Band-pass; clip at $\pm 800\,\mu$V; recording-wise z-score	Current-epoch ResNet, then TCN

For E6, mean and population standard deviation (ddof = 0) are calculated over the entire filtered and clipped recording, including portions outside the later evaluation window. The implementation rejects zero or non-finite standard deviation rather than adding an epsilon; non-finite input/output signals are also rejected. Processing uses float64 for filtering and normalisation and stores model inputs as float32. On SHHS1 this is label-free target-record normalisation with a transductive dependency; it does not update model weights. E3 and E6 are trained separately on their respective source input variants, so their comparison is not an amplitude-only intervention on one frozen classifier.

The $\pm 800\,\mu$V clip is inactive in both cohorts: after band-pass filtering, no Sleep-EDF value exceeded $246\,\mu$V, every holdout SHHS1 EEG channel has a physical range of $\pm 125\,\mu$V, and the preprocessing audit of the first SHHS1 sample recorded no clipped samples. The first ResNet-1D convolution has no bias and is followed by batch normalisation, so dividing the input by 100 is equivalent to multiplying the stabilising constant ϵ of that layer by 10^4 when using batch statistics. Thus scale invariance is approximate at finite ϵ; running-statistic initialisation and numerical precision can also affect the fitted models (Supplementary Section S4). For E6, Sleep-EDF recordings last 17–24 h (median 23 h) and include long daytime segments outside the evaluation window, whereas SHHS1 recordings cover the night only (holdout sample: 5.5–9.1 h, median 8.8 h). The per-recording standard deviation of the E6 input within the evaluation window therefore has a median of 0.77 on Sleep-EDF but about 0.99 on SHHS1 (Supplementary Section S3, S6).

The evaluation window extends up to 30 minutes before the first and after the last reference sleep epoch (N1, N2, N3 or REM), clipped to recording limits. Original stage 3/4 labels are merged as N3; unknown/movement epochs are unscored. This reference-dependent trimming convention and the later boundary diagnostics must be distinguished from label-free model fitting/inference. Sleep-EDF metrics use out-of-fold predictions, with each subject evaluated by a model not trained on that subject. SHHS1 uses the arithmetic mean of class probabilities from all ten source-fold models before argmax. Cross-cohort score differences therefore compare distinct cohorts and evaluation settings, not an isolated effect of dataset shift.

Runtime measurement

The archived forward-pass benchmark used the selected fold-00, seed-42 checkpoints on a Tesla V100-PCIE-16GB GPU, Python 3.10.13, PyTorch 2.5.1+cu124, CUDA 12.4 and cuDNN 9.1. It used float32 synthetic input of shape $(1, 100, 1, 3000)$, 20 warm-up passes and 100 timed repetitions in each of three rounds, with model order varied across rounds and GPU synchronisation around timing. Both encoder and sequence forward passes are included; disk I/O, preprocessing, feature-cache serialization and training are excluded. Latency is the median of the recorded passes; memory is the maximum peak allocated GPU memory, distinct from reserved or incremental memory. These measurements characterise offline computation, not the time until future EEG needed by a bidirectional model becomes available. Training-and-validation wall-clock times additionally depend on the recipes and stopping rules in Table S2.

Statistical implementation

Bootstrap samples draw subjects with replacement and use the same sampled subjects for both configurations, retaining all nights of each sampled subject. The 95% percentile interval uses the 2.5th and 97.5th percentiles of 10,000 paired resamples. The primary Sleep-EDF analysis uses bootstrap seed 2026; context analyses use 2028, and the SHHS1 E1/E2 and E3−E2 extensions use 2031 and 2032, respectively. Each analysis retains its archived seed and comparison family. Two-sided Wilcoxon tests use `zero_method="wilcox"`, excluding zero differences from ranking, no continuity correction and automatic selection of the calculation method. If all paired differences are zero, the implementation returns $p = 1$. Holm correction applies to Wilcoxon p-values within each stated family, not to the percentile intervals. Non-significance is not interpreted as equivalence between configurations.

Supplementary results

First SHHS1 sample: performance, N3 errors and ADAST budgets

The following three tables accompany Sections 4.4, 4.6 and 4.7 of the main text. On this sample, the unadjusted exact sign tests for the two primary comparisons, E3−E0 and E3−E6, gave $p = 3.8 \times 10^{-13}$ and 1.9×10^{-7}. Within ±1 epoch of a reference-label transition, the pooled boundary macro-F1 of E3 exceeded that of E0 by 0.0366 and that of E6 by 0.0183. For E6, N3 recall near reference-label changes was 0.0721, compared with 0.0733 for E3.

Table S5: Cross-cohort performance on the first SHHS1 sample (n = 180, seed 42). All model weights were trained on Sleep-EDF. E6 additionally uses complete-record, unlabelled target statistics. Intervals are subject-cluster bootstrap intervals for subject-mean macro-F1.

ID	**Subject-mean** macro-F1 **(95% CI)**	**Pooled** macro-F1	**Accuracy**	κ
E0	0.5268 [0.5101, 0.5431]	0.5761	0.6648	0.5339
E3	**0.5680 [0.5503, 0.5850]**	**0.6099**	**0.7016**	**0.5801**
E6	0.5407 [0.5227, 0.5584]	0.5732	0.6742	0.5408

Table S6: N3 under-detection on the first SHHS1 sample (seed 42). All models share 22,806 reference N3 epochs. The final column is the proportion of those epochs predicted as N2.

ID	N3 precision	N3 recall	N3 F1	N3→N2 count	Rate
E0	0.9007	0.2610	0.4048	16,480	72.3%
E3	0.9440	0.2582	0.4055	16,674	73.1%
E6	0.9052	0.2005	0.3283	17,764	77.9%

Table S7: ADAST learning-budget comparison on the first SHHS1 sample (180 participants; 169,012 scored epochs). All entries use epoch-30 checkpoints and ten-fold probability ensembles. Source-only is the matched ADAST control, not E3. Full source means 30 complete training passes.

Budget	Arm	Subject-mean macro-F1	Pooled macro-F1	N3 recall	N1 recall
1,140 updates	Source-only	.4949	.5539	.3195	.1974
1,140 updates	ADAST	.4497	.5072	.4269	.0000
Full source	Source-only	.5010	.5478	.2432	.1568
Full source	ADAST	.5455	.5918	.6111	.1057

Table S8: Per-class Sleep-EDF metrics for E3 and E6 (seed 42). Differences are calculated before rounding.

Stage	E3 Prec.	E3 Rec.	E3 F1	E6 Prec.	E6 Rec.	E6 F1	ΔF1
W	.9338	.9271	.9304	.9391	.9169	.9279	+.0026
N1	.5149	.5410	.5276	.5032	.5219	.5124	+.0152
N2	.8583	.8447	.8514	.8401	.8368	.8385	+.0130
N3	.8072	.8097	.8085	.7160	.7779	.7457	+.0628
REM	.8274	.8413	.8343	.8227	.8197	.8212	+.0131

Table S9: Pooled E3 confusion matrix on Sleep-EDF (rows: reference; columns: prediction).

	W	N1	N2	N3	REM
W	61,133	3,888	416	56	448
N1	3,283	11,643	4,867	105	1,624
N2	539	5,496	58,394	2,271	2,432
N3	30	45	2,377	10,558	29
REM	483	1,542	1,984	90	21,736

Table S10: Sleep-EDF pooled macro-F1 under two fixed seeds. Seed 42 is primary; seed 123 is a post-protocol sensitivity repeat. Differences are calculated before rounding.

Configuration	Seed 42	Seed 123	Difference
E0	.7754	.7755	+.0000
E1	.7802	.7776	-.0026
E2	.7835	.7828	-.0007
E3	.7904	.7883	-.0022
E4	.7891	.7887	-.0004
E6	.7691	.7780	+.0089

Table S11: Seed sensitivity of the protocol-specified Sleep-EDF contrasts. Bootstrap intervals describe differences in pooled macro-F1; Wilcoxon tests use paired subject-wise macro-F1. Holm correction is applied to the four tests within each seed, not to the marginal bootstrap intervals.

Contrast	Seed 42: Δ [95% CI], p_{Holm}	Seed 123: Δ [95% CI], p_{Holm}
E1−E0	.004811 [−.000915, .010193], .1022	.002188 [−.002870, .007043], .9130
E2−E1	.003251 [−.002370, .008808], .1036	.005143 [−.000571, .011292], .2400
E3−E2	.006962 [.000305, .014520], .8989	.005478 [−.002611, .015796], .9130
E3−E6	.021319 [.012179, .030698], .0012	.010249 [.002435, .017989], .1313

Table S12: Seed-123 wall-clock training and validation time (10 folds, sequential, same GPU). E1 time covers sequence training only because its encoder and cached features are reused from E0.

Configuration	Total	Mean per fold
E0: 15-CNN + BiLSTM	33 h 35 min 36 s	3 h 21 min 34 s
E1: 15-CNN + TCN	14 min 17 s	1 min 26 s
E2: ResNet-1D + TCN	2 h 44 min 57 s	16 min 30 s
E3: band-pass, clip, ÷100	3 h 16 min 40 s	19 min 40 s
E4: band-pass	2 h 55 min 56 s	17 min 36 s
E6: record-wise z-score	2 h 24 min 44 s	14 min 28 s

Table S13: Seed-123 sensitivity and E4 extension on the same 180 SHHS1 participants. These are post-protocol results on the previously evaluated cohort.

ID	Subject-mean macro-F1	Pooled macro-F1	Accuracy	κ
E0	.5303	.5785	.6738	.5420
E2	.5409	.5858	.6874	.5597
E3	.5634	.6031	.7003	.5772
E4	.5732	.6147	.7064	.5869
E6	.5503	.5865	.6873	.5552

Table S14: Paired SHHS1 seed-123 extension. Differences and subject-cluster percentile bootstrap intervals concern subject-mean macro-F1 (10,000 resamples). Two-sided Wilcoxon p-values are Holm-adjusted within this four-contrast extension, separately from the primary seed-42 family. W/T/L denotes subjects with a positive/zero/negative paired difference.

Contrast	Δ subject-mean macro-F1	95% CI	p_{Holm}	W/T/L
E4−E2	+.0323	[.0236, .0410]	7.40×10^{-13}	135/1/44
E3−E4	-.0098	[−.0125, −.0073]	1.89×10^{-10}	56/1/123
E3−E6	+.0131	[.0023, .0239]	.0085	111/0/69
E3−E0	+.0331	[.0225, .0438]	5.19×10^{-9}	131/0/49

Table S15: Additional E4−E3 comparisons on the same 180 SHHS1 participants and 169,012 labelled epochs. Every result uses the mean probabilities of all ten source-fold models at its stated seed. Intervals are paired subject-cluster percentile bootstrap intervals (10,000 resamples, bootstrap seed 2031), not intervals over training seeds. E4 seed 42 was newly inferred from historical checkpoints; its E3 control and both seed-123 results are verified archived predictions. The original seed-123 Holm analysis remains in Table S14; no new seed-42 p-value is added to the primary comparison family.

Seed	Model	Subject-mean macro-F1	Pooled macro-F1	N3 recall	N3 F1
42	E3	.5680	.6099	.2582	.4055
42	E4	.5811	.6234	.2728	.4233
123	E3	.5634	.6031	.2410	.3840
123	E4	.5732	.6147	.2591	.4068
E4−E3, seed 42		+.0131	95% CI [.0111, .0153] for subject-mean difference		
E4−E3, seed 123		+.0098	95% CI [.0073, .0125] for subject-mean difference		

Table S16: Secondary and post-hoc SHHS1 paired contrasts (seed 42; 180 subjects). Intervals are marginal 95% paired subject-cluster percentile bootstrap intervals for the subject-mean difference, with 10,000 resamples. The E1/E2 extension was specified before their SHHS1 inference but after the cohort had been evaluated with E0/E3/E6. E3−E2 was specified after their separate marginal results had been examined.

Contrast	Δ subject-mean macro-F1	95% CI	p	W/T/L
E1−E0	+.00651	[.00192, .01087]	.00325[a]	108/0/72
E2−E1	-.01280	[−.02209, −.00350]	.01005[a]	80/0/100
E3−E2	+.04750	[.03724, .05792]	2.35×10^{-17} [b]	147/0/33

[a] Two-sided subject-wise Wilcoxon, Holm-adjusted within the two-contrast E1/E2 extension (bootstrap seed 2031). [b] Unadjusted two-sided Wilcoxon for one separately specified post-hoc comparison (bootstrap seed 2032); it is not part of either historical Holm family. The contrasts compare specified pipeline configurations, not isolated architectural effects.

Table S17: Per-class transfer of E3; Pred/True is the predicted-to-reference class count ratio, not a measure of probability calibration. Differences are calculated before rounding.

	Sleep-EDF		SHHS1				
Stage	F1	Pred/True	Prec.	Rec.	F1	Pred/True	ΔF1
W	.9304	.993	.8823	.8014	.8399	.908	-.0905
N1	.5276	1.051	.2528	.5437	.3451	2.151	-.1824
N2	.8514	.984	.7394	.7349	.7372	.994	-.1143
N3	.8085	1.003	.9440	.2582	.4055	.273	-.4030
REM	.8343	1.017	.6052	.8939	.7218	1.477	-.1125

Table S18: Pooled E3 confusion matrix on SHHS1 (rows: reference; columns: prediction).

	W	N1	N2	N3	REM
W	29,638	4,481	1,171	21	1,672
N1	874	3,807	601	3	1,717
N2	2,348	5,947	56,063	324	11,605
N3	78	39	16,674	5,888	127
REM	655	784	1,311	1	23,183

Table S19: Per-class SHHS1 metrics for E3 and E2 (seed 42). Differences are calculated before rounding.

Stage	E3 Prec.	E3 Rec.	E3 F1	E2 Prec.	E2 Rec.	E2 F1	ΔF1
W	.8823	.8014	.8399	.8780	.7275	.7957	+.0442
N1	.2528	.5437	.3451	.2367	.3528	.2833	+.0618
N2	.7394	.7349	.7372	.7362	.6937	.7143	+.0228
N3	.9440	.2582	.4055	.9659	.2496	.3967	+.0087
REM	.6052	.8939	.7218	.4886	.9450	.6442	+.0776

Table S20: Reference-label oracle on E3 predictions (seed 42). Each row corrects only the indicated confusion-matrix cell. The percentage is relative to the observed pooled-score difference between Sleep-EDF out-of-fold evaluation and SHHS1 ensemble evaluation, not an isolated domain-shift effect or an estimate of achievable model performance.

Condition	macro-F1	Numerical gap recovered
Observed E3 on SHHS1	.6099	n/a
Correct N3→N2 only	.7444	.1345 (74.5%)
Correct N2→REM only	.6596	.0497 (27.5%)
E3 on Sleep-EDF	.7904	n/a

Table S21: Post-hoc reference-label boundary and stable-interior diagnostics. Seed 123 reuses the same subjects and label-defined regions; it is a sensitivity repeat, not an independent cohort. All metrics are pooled within each region.

Cohort	Seed	ID	Boundary macro-F1	Stable macro-F1	Boundary N3 recall	Stable N3 recall
Sleep-EDF	42	E0	.6179	.8175	.6121	.9216
Sleep-EDF	42	E3	.6321	.8346	.6326	.9316
Sleep-EDF	123	E0	.6181	.8173	.6174	.9162
Sleep-EDF	123	E3	.6343	.8299	.6385	.9285
SHHS1	42	E0	.4323	.5959	.0633	.4008
SHHS1	42	E3	.4689	.6220	.0733	.3900
SHHS1	123	E0	.4336	.5957	.0639	.3885
SHHS1	123	E3	.4605	.6166	.0656	.3660

For a label change between consecutive valid epochs $j-1$ and j, where j is the first epoch with the new label, the boundary neighbourhood contains $\{j-1, j, j+1\}$ within the same contiguous recording segment. Overlapping neighbourhoods are combined. Unknown-label gaps and recording boundaries break contiguity. Stable-interior epochs are at least three epoch positions from *both* sides of every label change, using distance $\min(|i-(j-1)|, |i-j|)$; segments without a change are stable. The two regions do not exhaust the scored epochs: Sleep-EDF contains 43,237 boundary, 130,693 stable and 21,539 remaining epochs; SHHS1 contains 44,744, 103,195 and 21,073, respectively. The remaining epochs are not simply the distance-two bin, because the boundary neighbourhood is centred on the first new-label epoch. These masks include all stage changes, not just N2-N3. Where persistent boundaries are reported in the archived diagnostics, each side must contain at least three consecutive epochs of its respective label. Region metrics retain the fixed five-class convention used in the main analysis.

E6 N3 recall is 0.2005 overall and 0.0721 in the same SHHS1 boundary neighbourhood (seed 42), compared with 0.2582 and 0.0733 for E3. These regions use reference labels on both sides and are offline diagnostics, not direct measurements of physiological transitions or deployable detectors. Differences in class composition between regions also preclude interpreting their score difference as an isolated effect of proximity to a label change. N3 under-detection persists in SHHS1 stable interiors in both seeds.

Supplementary analyses: representation and context

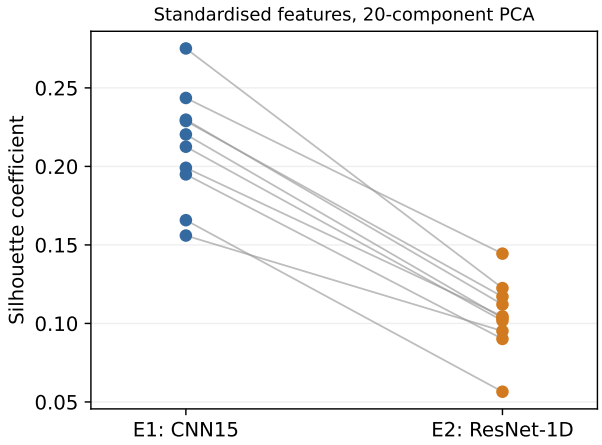
The representation analysis uses the same 1,000 held-out epochs per fold for E1 and E2: 200 per stage, sampled without replacement with a round-robin allocation across available subjects (sampling seed 42 + fold index). In each representation and fold separately, features are standardised and projected onto 20 principal components; these transforms are fitted to the sampled test features solely for this descriptive analysis, not fed back into the staging model. Silhouette uses Euclidean distance in the resulting space and the reference stage labels. The mean coefficient across ten folds is 0.2126 (SD 0.0356) for E1 and 0.1048 (SD 0.0229) for E2. E1 consists of 75 softmax-output features, whereas E2 is an unconstrained 128-dimensional representation; this difference and the analysis transforms prevent treating silhouette as a like-for-like measure of downstream predictive utility.

For the context ablation, the pretrained E0 encoder and 75-dimensional input interface are retained. Each removed feature dimension is replaced by its mean over valid labelled training epochs in that fold; the same replacement vector is applied to training, validation and test. TCN models are retrained for C, CP and CN with the specified E1 recipe and seed 42, while full CPN reuses E1. Thus the intervention changes the information available during both fitting and inference; it is not an inference-only deletion. The local protocol specifies the primary CPN-C boundary contrast and two secondary contrasts, with Holm correction across those three subject-wise Wilcoxon tests. Bootstrap intervals describe paired differences in pooled boundary macro-F1 under 10,000 subject-cluster resamples (seed 2028).

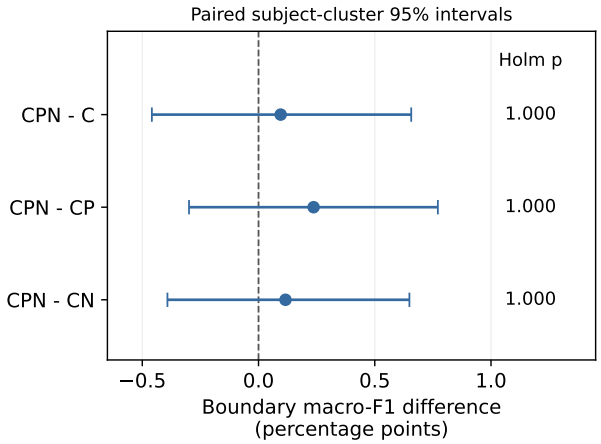
Table S22: Context-group ablation within E1 (seed 42). C, P and N denote current, previous and next epoch feature groups. Differences are calculated before rounding.

Condition	Overall macro-F1	Δ vs C	Boundary macro-F1	Δ vs C	N1 F1
C	.777477	n/a	.619102	n/a	.5011
CP	.777733	+.000257	.617686	-.001416	.5022
CN	.781585	+.004108	.618895	-.000207	.5055
CPN	.780230	+.002753	.620055	+.000953	.5115

For the protocol-specified CPN-C boundary comparison, Δmacro-F1 = 0.000953, 95% CI [−0.004588, 0.006568] and Holm $p = 1.000$ (36/78 subjects improved). Secondary CPN-CP and CPN-CN intervals also include zero with Holm $p = 1.000$. These comparisons did not detect an incremental boundary-performance benefit from the explicit P/N feature groups under the E1 recipe. The TCN continues to use temporal context across the sequence in every condition.



(a) Silhouette by fold.



(b) Context-ablation effects.

Figure S1: Supporting representation and context analyses. (a) Silhouette coefficients by test fold. (b) Paired 95% confidence intervals for pooled boundary macro-F1 differences: CPN minus C, CP and CN (Sleep-EDF, E1, seed 42). All intervals include zero; Holm-adjusted Wilcoxon $p = 1.000$ for each contrast. Plots were drawn from archived aggregates using repository code with OpenAI Codex assistance.

Supplementary Methods S2: additional intervention experiments

These experiments address errors observed in the original SHHS1 evaluation. They use the same 180 test participants. Model selection uses source validation, and adaptation uses unlabelled target data. All target recordings are processed in full, without annotation-based windowing, before applying the historical scoring mask (169,012 labelled epochs from 183,528 inferred epochs). Every contrast uses a control with the same inference context. In particular, the unmodified full-record E3 control is not interchangeable with the historical trimmed-context E3 ensemble. We report 10,000 paired participant-bootstrap resamples (seed 2031), with marginal 95% intervals, separately from the primary Holm families. These intervals condition on the fitted models and do not incorporate training variability.

Temperature scaling and prior correction: ten-fold ensemble

For each seed-123 E3 fold, a positive temperature is fitted by minimising negative log-likelihood on that fold's archived source-validation predictions, with bounds $[0.05, 20]$. This validation set was also used for historical checkpoint selection. The source prior is the mean calibrated validation posterior. EM estimates a target prior from 4,989 full-record epochs of five fixed, unlabelled adaptation participants, disjoint from the 180 test participants. Initialisation uses the source prior; the stopping tolerance is 10^{-8}, the maximum is 1,000 iterations, and a 10^{-12} floor prevents zero probabilities. No true target prevalence or class-specific multiplier is supplied. For each fold, the calibrated posterior is multiplied by the estimated target/source prior ratio and renormalised. All ten fits are frozen before target inference. We average all ten fold probabilities separately for the raw, temperature-only and temperature-plus-EM arms, without fold selection. Temperature scaling preserves each model's argmax but can alter the argmax after probability averaging across differently calibrated models.

Table S23: Additional calibration experiment: all ten E3 folds, seed 123, full-record inference on 180 SHHS1 participants. Each column averages the same ten models. A dash denotes undefined N3 precision because the EM ensemble predicts no N3 epochs.

Metric	Raw	Temperature	Temperature + EM
Subject-mean macro-F1	.5590	.5595	.4899
Pooled macro-F1	.5989	.5994	.5021
Accuracy	.6957	.6957	.6372
N3 precision	.9443	.9446	n/a
N3 recall	.2410	.2421	.0000
N3 F1	.3840	.3855	.0000
N2 recall	.7306	.7287	.7288

Temperature-only minus raw subject-mean macro-F1 is 0.00049 (95% CI $[0.00002, 0.00097]$), a small absolute change. Temperature-plus-EM minus temperature-only is -0.0696 ($[-0.0793, -0.0600]$). All ten EM fits converge, but each places its N3 prior near the numerical floor; 98.83% of reference N3 epochs are then labelled N2. The estimated prior describes the output of this EM procedure on five adaptation participants, rather than their measured N3 prevalence. In this experiment, EM correction eliminated N3 predictions and reduced subject-mean macro-F1 relative to temperature scaling alone.

Class-weighted TCN: matched ten-fold ensemble

For each seed-123 fold, the corresponding verified historical E3 encoder is frozen and both sequence classifiers are trained with the same initial weights and source batch order. The control uses unweighted cross-entropy; the intervention uses $N/(5n_c)$ computed separately from that fold's valid source-training labels. Both arms use Adam at learning rate 0.0005, batches of eight recordings, gradient norm clipping at 1, at most 300 epochs and patience 30. Strict maximum pooled source-validation macro-F1 selects each checkpoint. No target labels are used for fitting, stopping, class weights or selection. All 20 checkpoints are frozen before the new target ensemble evaluation, and all ten probabilities are averaged in each arm. The historical E3 model is not the matched unweighted control.

Verified CPU pairs from folds 0–5 are reused; folds 6–9 are fresh CUDA pairs, not continuations of partially trained CPU models. Both arms within each fold use the same backend. PyTorch versions are 2.5.1 (CPU) and 2.11.0 (CUDA); the experiment is therefore mixed-backend, not a uniform-backend replication. Independent CPU replay of CUDA source-validation and outer-test predictions reproduces their confusion matrices and metrics. New training timings do not replace the historical operational benchmark. This intervention trains sequence classifiers on frozen features, not the encoders end-to-end. The earlier fold-0 result is retained below, not counted as an independent replication of the ten-fold result.

Table S24: Matched class-weighting experiment: ten-fold ensembles, seed 123, full-record inference on 180 SHHS1 participants. Both arms average ten newly trained sequence classifiers with the same fold-specific frozen encoders. Per-class metrics and error rates are pooled.

Metric	Unweighted	Weighted
Subject-mean macro-F1	.5530	.5514
Pooled macro-F1	.5930	.5928
Accuracy	.6892	.6663
N3 precision	.9462	.9342
N3 recall	.2345	.3225
N3 F1	.3759	.4795
N2 recall	.7260	.6728
N3→N2 rate	.7497	.6549
N2→N3 rate	.0038	.0063

Weighted minus unweighted subject-mean macro-F1 is -0.0016 (95% CI $[-0.0049, 0.0017]$). For pooled metrics, the differences are -0.0002 ($[-0.0032, 0.0027]$) in macro-F1, 0.1036 ($[0.0938, 0.1137]$) in N3 F1, 0.0880 ($[0.0805, 0.0954]$) in N3 recall, -0.0120 ($[-0.0187, -0.0050]$) in N3 precision and -0.0531 ($[-0.0577, -0.0488]$) in N2 recall. N3→N2 errors decrease by 0.0948, while N2→N3 errors increase by 0.0025. Intervals that include zero do not establish equivalence or non-inferiority. Bootstrap resampling pairs participants, not epochs, and conditions on the fixed fitted models; it does not quantify training or backend variability.

On the source outer tests (78 participants, 195,469 labelled epochs, out-of-fold rather than ensembled predictions), unweighted/weighted pooled macro-F1 is 0.7878/0.7840 and accuracy 0.8330/0.8213. N3 recall increases from 0.8059 to 0.8724, whereas precision decreases from 0.8121 to 0.7404 and F1 from 0.8090 to 0.8010. Increased recall alone therefore does not establish an overall improvement on source or target.

Class-weighted TCN: earlier matched single-fold pilot

This comparison uses fold 0 and seed 123. The verified E3 encoder is frozen, and both TCNs are trained anew with identical initial weights and source batch order. The control uses unweighted cross-entropy; the intervention uses $N/(5n_c)$ from valid source-training labels only. Weights in W/N1/N2/N3/REM order are $(0.5755, 1.8330, 0.5671, 3.2466, 1.5493)$. Both arms use Adam with learning rate 0.0005, batches of eight recordings, gradient norm clipping at 1, at most 300 epochs, and patience 30. Strict improvement in pooled source-validation macro-F1 determines selection. The unweighted/weighted runs stop at epochs 42/53 and select epochs 12/23. Target labels are not used for fitting, stopping, class-weight choice or checkpoint selection. The retrained unweighted TCN, not the historical E3 checkpoint, is the matched control.

Table S25: Matched class-weighting experiment: fold 0, seed 123, 180 SHHS1 participants. Both columns are single-model predictions, not ten-fold ensembles.

Metric	Unweighted	Weighted
Subject-mean macro-F1	.5431	.5363
Pooled macro-F1	.5861	.5790
Accuracy	.6837	.6510
N3 precision	.9400	.9253
N3 recall	.2432	.3221
N3 F1	.3864	.4779
N2 recall	.7646	.6708

Weighted minus unweighted subject-mean macro-F1 is -0.0068 (95% CI $[-0.0135, 0.0001]$), whereas N3 recall increases by 0.0789 ($[0.0720, 0.0863]$) and N2 recall decreases by 0.0938 (difference interval $[-0.1097, -0.0785]$). The first interval does not establish equivalence or non-inferiority. On the source outer-test set (eight participants, 19,506 labelled epochs), pooled macro-F1 is 0.7716/0.7672 and N3 recall is 0.8678/0.9098 for unweighted/weighted training. N3 F1 decreases from 0.8031 to 0.7698 on source, showing that greater recall alone is not sufficient evidence of a better classifier.

ADAST: limited-budget matched ten-fold adaptation comparison

Ten new CUDA pairs use the same harmonised ADAST design as the earlier pilot below, on all source outer folds at seed 123. The upstream model/configuration files and similarity penalty are unchanged. Both arms in every fold start from identical weights and receive the same source permutations and 1,140 updates. Each epoch samples 4,864 valid source-training epochs, not a complete pass through that fold's source data. The same five unlabelled adaptation participants supply 4,989 epochs; the 180 test participants provide no training signals or labels. All 20 final fixed-budget checkpoints are frozen before local target inference or scoring. For each arm, maximum dual-head logits are softmaxed within each fold and all ten probability arrays are averaged. Target inference uses source attention for the source-only arm and target attention for ADAST. The historical full-record scoring mask is then applied to both arms. Source results use each outer fold's test subjects once (78 participants, 195,469 valid epochs), with source attention for both arms; they are out-of-fold predictions, not ensembles. The CUDA pairs and CPU inference are kept separate from the historical operational benchmark. Participant-bootstrap intervals condition on these fitted models, not training randomness.

Table S26: Matched ten-fold ADAST comparison, seed 123: 180 SHHS1 participants, 169,012 valid scored epochs. Both arms use ten fresh CUDA-trained models and local full-record inference; the source-only arm is the matched dual-head control, not E3.

Metric	Source-only	ADAST
Subject-mean macro-F1	.4949	.4497
Pooled macro-F1	.5539	.5072
Accuracy	.6977	.6951
Cohen's κ	.5488	.5367
N3 precision	.9266	.9225
N3 recall	.3195	.4269
N3 F1	.4752	.5837
N1 recall	.1974	.0000
N2 recall	.8712	.8655
REM recall	.4343	.2938
N3→N2 rate	.6747	.5588

ADAST minus source-only subject-mean macro-F1 is -0.0452 (95% CI $[-0.0558, -0.0347]$), whereas the N3-recall difference is 0.1074 ($[0.0714, 0.1422]$) and the N3-F1 difference is 0.1085 ($[0.0721, 0.1442]$). N3→N2 errors decrease, but more than half of true N3 epochs remain misclassified as N2. The ADAST ensemble produces no N1 predictions, and

REM recall decreases. On source out-of-fold testing, source-only/ADAST pooled macro-F1 is 0.6665/0.5276, accuracy is 0.7724/0.7116, N3 recall is 0.7689/0.5933, N1 recall is 0.1538/0.0039 and REM recall is 0.5895/0.2337. The source subject-mean macro-F1 difference is −0.1388 ([−0.1520, −0.1250]). Thus, target N3 recall improved while aggregate source and target performance declined. The comparison uses a shared architecture and training budget for ADAST and its source-only control. The subsequent source-validation studies examine training-budget and loss-component effects; the full-source ten-fold evaluation below tests transfer under the larger budget.

ADAST: earlier matched single-fold pilot

The harmonised implementation uses the upstream ADAST model classes and similarity penalty at revision e0fb503, with this study's E3 preprocessing and source split (fold 0, seed 123). It is not an exact reproduction of the published data protocol. Both arms use the same initialisation, source permutations and 1,140 updates: two rounds of 15 epochs, each with 38 batches of 128 epochs. The shorter unlabelled-target loader determines this budget; each epoch presents 4,864 of the 157,200 source-training epochs, not a full source pass. The five adaptation participants supply 4,989 epochs. The matched source-only control retains the two classifiers and similarity regularisation, but has no target forward pass, adversarial loss or pseudo-label loss. Both arms use the same source-loss schedule (1 then 0.1), Adam (learning rate 0.001, betas 0.5/0.99, weight decay 0.0003). After epoch 10 of round 0, the learning rate is multiplied by 0.1 for every joint-optimizer parameter group in the implemented adapter, not only the encoder. ADAST uses adversarial weight 1, similarity weight 0.001, and target-loss weights 0 then 0.01. Pseudo-labels are fixed at each round's start from the mean of the two logits. Inference takes the elementwise maximum of those logits, with source attention for the control and target attention for ADAST on SHHS. The final fixed-budget checkpoints are used; no target-label validation or performance-based selection is performed.

Table S27: Harmonised ADAST comparison: fold 0, seed 123, 180 SHHS1 participants. The source-only arm is the matched dual-head control, not E3.

Metric	Source-only	ADAST
Subject-mean macro-F1	.4755	.4214
Pooled macro-F1	.5319	.4766
Accuracy	.6820	.6762
N3 precision	.9277	.9270
N3 recall	.2657	.2622
N3 F1	.4131	.4087
N1 recall	.2521	.0077

ADAST minus source-only subject-mean macro-F1 is −0.0542 (95% CI [−0.0641, −0.0443]); the N3-recall difference is −0.0035 ([−0.0426, 0.0345]). On the source outer-test set, evaluated with source attention in both arms, pooled macro-F1 is 0.6562/0.5015 and N1 recall is 0.1049/0.0015. Thus the deterioration is not restricted to SHHS. This single-fold, limited-budget comparison does not rank ADAST against the ten-fold E3 ensemble or establish that unsupervised adaptation is generally ineffective. This pilot is retained separately from the new ten-fold CUDA comparison. Subsequent source-validation budget and loss-component experiments are reported below.

ADAST: source-validation learning-budget and loss-component experiments

Two subsequent development campaigns used source fold 0, seed 123: source-only/ADAST at limited/full-source learning budgets, then a fresh full-source ADAST reference with three single-factor loss interventions. Training and source-validation data comprise 157,200 and 18,763 valid epochs respectively. Both campaigns reuse the same five unlabelled adaptation participants (4,989 epochs), without source outer-test or SHHS test access. Each model trains for two rounds of 15 epochs on T4 CUDA float32, using matched initialisation, source order and the earlier optimiser schedule. Limited training has 38 updates per epoch (1,140 total); full-source training has 1,229 updates per epoch (36,870 total), retaining the final batch of 16 source epochs. The larger budget also increases target presentations and changes scheduler timing in update units; it is not a source-coverage-only intervention.

Both attention paths are evaluated on the same source validation set. The selected checkpoint maximises source-attention validation macro-F1, retaining the first tie; epoch-30 checkpoints are reported separately. No target-attention validation rule or SHHS score selects a checkpoint. Full-validation CPU inference from all eight selected/final checkpoints in each campaign and recomputation of all 124 initial/per-epoch validation diagnostics are performed independently. Participant-bootstrap intervals from the ten-fold transfer experiments are not attached to these one-fold development comparisons.

Table S28: Source-validation learning-budget comparison: fold 0, seed 123. All scores use source attention; selected checkpoints follow the fixed validation criterion.

Arm	Budget	Selected epoch	Selected macro-F1	Selected N1 recall	Final macro-F1	Final N1 recall
Source-only	Limited	15	.6585	.1956	.6316	.1243
ADAST	Limited	11	.6470	.1673	.5236	.0000
Source-only	Full source	9	.7174	.5380	.6911	.2847
ADAST	Full source	22	.6934	.2338	.6784	.1591

The larger budget improves selected-checkpoint macro-F1 in both learning conditions and avoids complete N1 loss at epoch 30 in ADAST. Nevertheless, selected-checkpoint source-only macro-F1 remains higher. For the full-source pair at epoch 30, ADAST/source-only N3 recall is .8068/.5783, whereas N1 recall is .1591/.2847. In the limited ADAST arm, N1 recall falls .1060 → .0096 at epochs 15 → 16, when source CE weight falls 1 → .1 and pseudo CE weight rises 0 → .01 simultaneously. In the full-source ADAST reference, N1 recall instead rises .1660 → .1934 at that boundary. The limited-budget trajectory is not a universal epoch-16 collapse.

The second campaign keeps the full-source budget and changes one configured component per arm: source CE remains 1 in round two, pseudo CE remains 0 in both rounds, or adversarial weight is 0. The latter arm skips discriminator forward/backward/optimiser steps, retaining target forward passes, the pseudo-label schedule and other losses. The discriminator weights remain unchanged and its optimiser state remains empty. The reference, keep-source and no-pseudo arms have identical saved validation logits through epoch 15; their interventions take effect from epoch 16. The fresh reference reproduces all 30 validation metrics of the previous full-source ADAST arm. Similarity weight remains .001 throughout.

Table S29: Full-source ADAST loss interventions on source validation: fold 0, seed 123. Selected and final checkpoints are separate comparisons. The selected no-pseudo checkpoint precedes intervention activation and is also present in the reference trajectory.

Arm	Checkpoint	Epoch	macro-F1	N1 recall	N3 recall	REM recall
Reference	Selected	22	.6934	.2338	.8261	.7139
Keep source CE	Selected	22	.6941	.2664	.8323	.6490
No pseudo CE	Selected	7	.6891	.4902	.8834	.5486
No adversarial loss	Selected	29	.6943	.2416	.7115	.6865
Reference	Final	30	.6784	.1591	.8068	.7544
Keep source CE	Final	30	.6809	.2064	.8095	.6742
No pseudo CE	Final	30	.6717	.1769	.8192	.6430
No adversarial loss	Final	30	.6777	.2425	.6025	.6764

Retaining source CE increases selected-checkpoint N1 F1 .2988 → .3238 and recall .2338 → .2664, while N1 precision is .4138/.4128. N3 recall increases slightly but N3 precision decreases .7533 → .7399 and F1 .7880 → .7834; REM recall and F1 decrease. Its selected macro-F1 gain over the reference is .000673. Removing adversarial loss gives the highest selected macro-F1 among these four arms, a gain of .000887, but lowers N3 recall by .1146 and raises N3 precision .7533 → .8294. At fixed epoch 30, no-pseudo improves N1 recall modestly but lowers overall macro-F1 and REM recall; no-adversarial increases N1 recall but lowers N3 recall by .2043. High N1 recall at the selected no-pseudo epoch 7 is not an effect of disabling pseudo-label learning, which has zero weight in the reference at that point. These interventions identify distinct stage-wise trade-offs rather than a uniformly improved configuration.

Table S30: Five-stage F1 for the full-source ADAST loss interventions on source validation.

Arm	Checkpoint	W	N1	N2	N3	REM
Reference	Selected	.8771	.2988	.8151	.7880	.6880
Keep source CE	Selected	.8831	.3238	.8100	.7834	.6700
No pseudo CE	Selected	.8818	.4055	.7692	.7776	.6116
No adversarial loss	Selected	.8804	.3111	.8205	.7660	.6934
Reference	Final	.8812	.2295	.8073	.7944	.6796
Keep source CE	Final	.8835	.2722	.8014	.7880	.6596
No pseudo CE	Final	.8797	.2465	.7925	.7848	.6548
No adversarial loss	Final	.8772	.3074	.8108	.7173	.6757

Table S31: Target-attention diagnostics of the same checkpoints on the same source validation set, not SHHS evaluation. Target attention does not select checkpoints.

Arm	Checkpoint	macro-F1	N1 recall	N3 recall
Reference	Selected	.6780	.2225	.8302
Keep source CE	Selected	.6789	.2599	.8599
No pseudo CE	Selected	.6803	.4259	.8551
No adversarial loss	Selected	.5227	.1521	.0773
Reference	Final	.6674	.1586	.8116
Keep source CE	Final	.6628	.1899	.8682
No pseudo CE	Final	.6546	.1517	.8116
No adversarial loss	Final	.5026	.1352	.0628

Without adversarial loss, the target-attention path receives neither adversarial nor pseudo-CE loss contributions in round one; source CE uses source attention, and similarity regularises the two classifier heads. Target forward passes and optimiser/weight decay are retained. The no-adversarial path therefore has a different learning condition, as intended by the intervention. Its poor target-attention N3 recall in Table S31 must be considered alongside the stronger source-attention scores; it is not evidence of better cross-cohort adaptation.

The selected source-validation ranking is no-adversarial, keep-source, reference, then no-pseudo by selected validation macro-F1; all remain below the full-source source-only score .7174. These source-validation comparisons distinguish the effects of loss components from the ten-fold SHHS transfer results reported above.

ADAST: three-arm confirmation on another source fold

Fold 1, seed 123, compared fresh source-only, ADAST-reference and retained-source-CE models with identical source orders; the ADAST arms shared target orders and round-one predictions. Each arm used 30 full-source passes over 152,825 training epochs: 1,194 updates per pass (batch 128, final batch 121), or 35,820 updates. Validation comprised 23,881 epochs (W/N1/N2/N3/REM: 9,912/1,911/7,133/1,968/2,957). The same five unlabelled adaptation participants were reused, without source outer-test or SHHS test access. Selection retained the source-validation rule above; epoch 30 was reported separately.

The comparison reports aggregate macro-F1 and all five stage F1 scores for both attention paths, using source attention for checkpoint selection. Selected and final checkpoints are reported separately to distinguish the best validation result from the end-of-training result.

Table S32: Fold-1 confirmation, seed 123: selected and final source-attention metrics on the same source validation. Stage columns are F1.

Arm	Checkpoint	Epoch	macro-F1	W	N1	N2	N3	REM
Source-only	Selected	10	.7370	.9432	.3200	.8427	.8568	.7224
Reference	Selected	27	.6907	.9392	.2175	.8021	.8096	.6852
Keep source CE	Selected	27	.6892	.9382	.2302	.7935	.8161	.6680
Source-only	Final	30	.7176	.9444	.2415	.8333	.8503	.7184
Reference	Final	30	.6857	.9391	.2175	.7968	.8077	.6672
Keep source CE	Final	30	.6814	.9397	.2158	.7865	.7999	.6653

Table S33: Target-attention diagnostics for the fold-1 checkpoints on the same source validation, not SHHS. This path does not select checkpoints. Stage columns are F1.

Arm	Checkpoint	macro-F1	W	N1	N2	N3	REM
Source-only	Selected	.6914	.8955	.2911	.8201	.7880	.6624
Reference	Selected	.6817	.9339	.2089	.7937	.8003	.6715
Keep source CE	Selected	.6727	.9318	.2085	.7802	.7864	.6564
Source-only	Final	.6213	.8318	.2052	.7650	.6403	.6642
Reference	Final	.6824	.9360	.2115	.7949	.7951	.6743
Keep source CE	Final	.6714	.9343	.2073	.7761	.7731	.6660

Retaining source loss increased selected N1 recall .1502 → .1627, but reduced REM recall .7376 → .6980. N3 recall changed .9644 → .9649 and precision .6975 → .7070. Through target attention, N3 recall increased .9670 → .9756 but precision decreased .6826 → .6587, lowering F1. At epoch 30, N1 F1 and macro-F1 were also lower than the reference: the

selected N1 benefit was not uniform across checkpoints or paths. The fold-0 N1 gain and REM trade-off remain separate. Training/validation used 2,975.78 seconds on T4, excluding upload and result verification.

ADAST: full-source ten-fold transfer and matched budget contrast

The full-source comparison comprises ten source-only/ADAST pairs at seed 123. Four previously evaluated models from folds 0–1 are reused after checking their checkpoint, input and verification hashes; the retained-source-CE and other ablation arms are excluded. Sixteen models for folds 2–9 complete 30 epochs on T4 CUDA. Each epoch covers the entire source-training partition for that fold, yielding 35,670–37,620 updates per model over 30 epochs. The paired arms share initial weights and source permutations and use the same five unlabelled adaptation participants (4,989 epochs). The input bundle physically contains the source records covered by the earlier authorised fold-0/1 train/validation union; each model uses only its own fold's training and validation roles. SHHS evaluation signals and labels remain local and are not used for training.

The main comparison uses epoch 30 in both arms and both budgets, rather than choosing a checkpoint on SHHS. Within each fold, the maximum of the two classifier logits is softmaxed; probabilities are averaged in float64 across ten folds before float32 storage. SHHS uses source attention for source-only and target attention for ADAST. The full recordings supply 183,528 input epochs and the common scoring mask retains 169,012 epochs from 180 participants. Sleep-EDF outer-test predictions use source attention for both arms, each participant in its own fold; these are out-of-fold results, not ensemble predictions.

The new training outputs pass independent recomputation of all 496 validation diagnostics and CPU inference of all 32 selected/final checkpoints on their complete source-validation partitions through both attention paths. The local evaluation recomputes prediction hashes, confusion matrices and participant bootstrap results, with 40 target model-record and 20 source model-batch inference replays. These checks supplement the proof checks for the four reused models.

Table S34: Epoch-30 full-source pooled class metrics. SHHS1: 180 participants, 169,012 scored epochs, ten-model ensemble per arm. Sleep-EDF: 78 participants, 195,469 outer-test epochs, out-of-fold predictions. SO is the matched source-only dual-head control; AD is ADAST. Supports are true-class epoch counts.

Cohort	Arm	Stage	Precision	Recall	F1	Support
SHHS1	SO	W	.8317	.8372	.8344	36,983
	SO	N1	.1267	.1568	.1401	7,002
	SO	N2	.7152	.7981	.7544	76,287
	SO	N3	.9463	.2432	.3869	22,806
	SO	REM	.5631	.6976	.6231	25,934
	AD	W	.8216	.8441	.8327	36,983
	AD	N1	.1299	.1057	.1165	7,002
	AD	N2	.8103	.6369	.7132	76,287
	AD	N3	.8823	.6111	.7221	22,806
	AD	REM	.4375	.8360	.5744	25,934
Sleep-EDF	SO	W	.9061	.9293	.9176	65,941
	SO	N1	.4392	.2850	.3457	21,522
	SO	N2	.7892	.8878	.8356	69,132
	SO	N3	.8209	.7868	.8035	13,039
	SO	REM	.7362	.6727	.7030	25,835
	AD	W	.9216	.9034	.9124	65,941
	AD	N1	.4374	.2255	.2976	21,522
	AD	N2	.7580	.8429	.7982	69,132
	AD	N3	.6864	.8995	.7786	13,039
	AD	REM	.6295	.6278	.6287	25,835

Table S35: Matched participant-bootstrap budget contrasts on the same 180 SHHS1 participants. Effects are ADAST minus its source-only control; the difference of effects is full-source minus limited-budget. All four systems use the same 10,000 participant resamples (seed 2031). Intervals condition on the fitted models.

Contrast / metric	Difference	95% CI
Limited: subject-mean macro-F1	−.0452	[−.0558, −.0347]
Full-source: subject-mean macro-F1	.0445	[.0352, .0539]
Difference of effects: subject-mean macro-F1	.0897	[.0767, .1023]
Full-source: pooled macro-F1	.0440	[.0345, .0536]
Full-source: accuracy	−.0025	[−.0140, .0088]
Full-source: N1 F1	−.0236	[−.0385, −.0077]
Full-source: N2 F1	−.0412	[−.0549, −.0281]
Full-source: N3 F1	.3351	[.3025, .3683]
Full-source: REM F1	−.0488	[−.0629, −.0339]
Full-source: N3 recall	.3679	[.3442, .3912]
Full-source: N3→N2 rate	−.3836	[−.4076, −.3590]
Full-source: N2→N3 rate	.0192	[.0146, .0245]

Full-source subject-mean macro-F1 is .5010/.5455 and pooled macro-F1 .5478/.5918 for source-only/ADAST. N3→N2 errors fall from 17,163 to 8,414 of 22,806 true N3 epochs (rates .7526/.3689), while N2→N3 errors rise from 299 to 1,763 of 76,287 true N2 epochs (.0039/.0231). The N3 benefit is accompanied by lower N1, N2 and REM F1. REM recall increases, but its precision decreases enough to lower F1; recall alone therefore does not demonstrate a general REM improvement. Accuracy changes .6898 → .6873, with a paired interval spanning zero, not evidence of equivalence.

On Sleep-EDF, source-only/ADAST subject-mean macro-F1 is .6613/.6281 (difference −.0332, 95% CI [−.0434, −.0229]), pooled macro-F1 .7211/.6831 and accuracy .8003/.7707. N3 recall increases while its precision and F1 decrease. Thus the positive SHHS class-balanced effect does not represent uniform improvement across classes or cohorts. Increasing the budget also increases target exposure and changes the scheduler's timing in update units; the contrast does not isolate source-data coverage. Folds and epochs are not treated as independent statistical samples, and the intervals do not include seed-to-seed training variation. The historical latency and workflow benchmarks are unchanged by this CUDA campaign.

ADAST: secondary source-validation checkpoint evaluation

Each of the same twenty full-source models supplies its checkpoint with the highest source-validation macro-F1 through source attention, using the first epoch in a tie. The SHHS attention paths, ten-fold probability averaging and scoring mask are unchanged; source results remain out-of-fold through source attention. No model is retrained or selected by SHHS performance. This secondary analysis changes the checkpoint rule, not the participants, adaptation data, loss or training budget. Epoch 30 remains the primary comparison across the two learning budgets.

Table S36: Secondary full-source evaluation with checkpoints selected by source validation. SHHS1 uses ten-fold ensembles on the same 180 participants; Sleep-EDF uses OOF predictions for 78 participants. Differences are ADAST minus source-only. The 10,000 participant-bootstrap draws (seed 2031) condition on fitted models.

Cohort	Arm	Subject-mean macro-F1	Pooled macro-F1	Accuracy
SHHS1	Source-only	.5073	.5552	.6818
	ADAST	.5396	.5854	.6709
Sleep-EDF	Source-only	.6681	.7237	.7947
	ADAST	.6368	.6921	.7730

Cohort	Contrast metric	Difference	95% CI
SHHS1	Subject-mean macro-F1	.0322	[.0234, .0412]
	Pooled macro-F1	.0302	[.0212, .0395]
	N3 recall	.3267	[.3050, .3483]
	N3 F1	.2892	[.2590, .3200]
Sleep-EDF	Subject-mean macro-F1	−.0313	[−.0416, −.0205]
	Pooled macro-F1	−.0316	[−.0416, −.0221]

Table S37: Secondary full-source pooled class metrics at source-validation-selected checkpoints. SHHS1: 169,012 scored epochs; Sleep-EDF: 195,469 outer-test epochs. SO is source-only; AD is ADAST. Supports are true-class epoch counts.

Cohort	Arm	Stage	Precision	Recall	F1	Support
SHHS1	SO	W	.8367	.8253	.8310	36,983
	SO	N1	.1276	.1807	.1495	7,002
	SO	N2	.7278	.7555	.7414	76,287
	SO	N3	.9333	.2828	.4340	22,806
	SO	REM	.5302	.7466	.6201	25,934
	AD	W	.8256	.8374	.8314	36,983
	AD	N1	.1271	.1231	.1251	7,002
	AD	N2	.8112	.5992	.6893	76,287
	AD	N3	.8891	.6095	.7232	22,806
	AD	REM	.4163	.8464	.5581	25,934
Sleep-EDF	SO	W	.9189	.9120	.9154	65,941
	SO	N1	.4197	.3546	.3844	21,522
	SO	N2	.8090	.8580	.8328	69,132
	SO	N3	.7565	.8182	.7861	13,039
	SO	REM	.7200	.6805	.6997	25,835
	AD	W	.9231	.9024	.9126	65,941
	AD	N1	.4229	.2605	.3224	21,522
	AD	N2	.7735	.8341	.8026	69,132
	AD	N3	.7062	.8875	.7865	13,039
	AD	REM	.6248	.6485	.6365	25,835

For source-only/ADAST on SHHS1, N3→N2 errors are 16,214/8,290 of 22,806 true N3 epochs (rates .7110/.3635); N2→N3 errors are 383/1,638 of 76,287 true N2 epochs (.0050/.0215). N1, N2 and REM F1 decrease despite the positive aggregate class-balanced effect. The source outer-test loss also persists. Both checkpoint rules therefore show an N3 benefit with stage/cohort costs. They reuse the same cohorts and related fitted models, not independent replication samples.

Supplementary Section S3: confirmatory holdout evaluation

Protocol and sample

The protocol was committed to the project repository before any holdout recording was preprocessed. Commit timestamps are recorded by the authors' machine; as an independent record, the GitHub server logged an automated test run on the protocol commit `a275617` at 11:18:09 UTC on 10 October 2026 (GitHub Actions run 38047923110), two days before the analysis script was committed at `2019b51`. It consists of the document `docs/SHHS_HOLDOUT_CONFIRMATORY_PROTOCOL_V1.md` and the configuration file `configs/shhs_holdout_confirmatory_v1.json`. Participants ranked 221–720 in the SHA-256 ranking used for the first SHHS1 sample formed the holdout sample; ranks 1–220 reproduced the roles of the first selection exactly. One ranked participant (rank 490) had no recording available from NSRR and was replaced, before any processing, by the participant at rank 721. All 500 recordings passed the technical audit (EDF and annotation hashes, channel, sampling rate and duration) without further replacement. The final sample comprised 500 participants (249 men and 251 women; age 62.8 ± 11.5 years, range 40–90) and 458,409 scored epochs in the evaluation windows.

Twelve arms were evaluated with locked checkpoints: E0, E2, E3, E4 and E6 at seeds 42 and 123, and the full-source ADAST and source-only arms (epoch-30 checkpoints). Each arm averaged the softmax probabilities of its ten source-fold models in float64 before argmax. Before inference, every checkpoint was restored from the audited repository history or from a verified backup and checked against its recorded SHA-256 value. No metric was computed before all arms had finished. The analysis script was committed before it was run and was run once.

The confirmatory family comprised H1 (E3−E0), H2 (E3−E6) and H3 (E3−E2) at seed 42. A hypothesis was confirmed if its Holm-adjusted two-sided Wilcoxon p-value was below 0.05 and its paired subject-cluster bootstrap interval (10,000 resamples, seed 2040) lay entirely above zero. The primary metric was subject-mean macro-F1 over five fixed classes, with the F1 of an absent class set to 0, as for the first sample. The protocol pre-specified seven secondary analyses (S1–S7), reported below. All intervals condition on the trained checkpoints and do not include training variability.

Table S38: All evaluated arms on the holdout SHHS1 sample (n = 500; 458,409 scored epochs). Subject-mean macro-F1 intervals are subject-cluster bootstrap intervals; the other columns are pooled over all epochs. ADAST and source-only are the full-source epoch-30 arms.

Arm	Seed	Subject-mean macro-F1 (95% CI)	Pooled macro-F1	Accuracy	κ	N3 recall	N3→N2
E0	42	0.5170 [0.5060, 0.5281]	0.5612	0.6506	0.5129	0.2565	72.3%
E2	42	0.5122 [0.5001, 0.5241]	0.5533	0.6512	0.5127	0.2484	71.5%
E3	42	0.5637 [0.5523, 0.5753]	0.6030	0.6957	0.5710	0.2549	72.7%
E4	42	0.5776 [0.5662, 0.5893]	0.6187	0.7103	0.5896	0.2742	71.1%
E6	42	0.5344 [0.5231, 0.5460]	0.5711	0.6740	0.5389	0.1987	77.1%
E0	123	0.5216 [0.5106, 0.5328]	0.5658	0.6625	0.5243	0.2506	73.2%
E2	123	0.5321 [0.5201, 0.5438]	0.5724	0.6737	0.5405	0.2402	72.1%
E3	123	0.5591 [0.5475, 0.5709]	0.5980	0.6965	0.5708	0.2392	74.1%
E4	123	0.5682 [0.5568, 0.5798]	0.6082	0.7013	0.5788	0.2574	72.6%
E6	123	0.5462 [0.5350, 0.5577]	0.5872	0.6876	0.5549	0.2018	77.2%
Source-only	n/a	0.4986 [0.4901, 0.5072]	0.5416	0.6844	0.5403	0.2366	75.3%
ADAST	n/a	0.5415 [0.5310, 0.5521]	0.5852	0.6798	0.5625	0.6161	35.8%

Table S39: Confirmatory family (seed 42) and its seed-123 replication (S1), each with its own Holm family of three. Δ is the difference in subject-mean macro-F1; median is the median per-participant difference; pooled columns give observed differences in pooled metrics.

	Contrast	Δ	95% CI	p_{Holm}	W/T/L	Median	Pooled macro-F1	Pooled κ
H1	E3−E0	0.0467	[0.0403, 0.0531]	2.0×10^{-37}	380/0/120	0.0398	0.0418	0.0581
H2	E3−E6	0.0293	[0.0231, 0.0354]	2.8×10^{-22}	342/0/158	0.0282	0.0319	0.0322
H3	E3−E2	0.0515	[0.0460, 0.0571]	1.2×10^{-50}	395/1/104	0.0400	0.0497	0.0583
S1	E3−E0	0.0375	[0.0307, 0.0444]	1.1×10^{-22}	345/0/155	0.0341	0.0322	0.0466
S1	E3−E6	0.0129	[0.0057, 0.0199]	1.7×10^{-5}	285/0/215	0.0119	0.0109	0.0159
S1	E3−E2	0.0270	[0.0217, 0.0325]	1.8×10^{-18}	327/1/172	0.0167	0.0256	0.0303

Secondary analyses

S2: E4−E3 and between-seed differences. The protocol expected E4−E3 not to exceed the between-seed differences of the same configuration systematically, and stated that E4−E3 would not be interpreted as an effect of preprocessing. This expectation was met at seed 123 but not at seed 42, where E4−E3 (0.0139) exceeded the absolute value of each between-seed difference (largest 0.0118, for E6). The between-seed difference of E2, which was not part of S2, was −0.0199 (seed 42 minus seed 123).

Table S40: S2: E4−E3 at each seed and between-seed differences of identical configurations (seed 42 minus seed 123). Intervals only; no hypothesis tests were pre-specified.

Contrast	Δ subject-mean macro-F1	95% CI	Δ pooled macro-F1	Δ accuracy
E4 (42)-E3 (42)	0.0139	[0.0126, 0.0153]	0.0157	0.0146
E4 (123)-E3 (123)	0.0091	[0.0074, 0.0108]	0.0102	0.0048
E3 (42)-E3 (123)	0.0046	[0.0026, 0.0067]	0.0050	−0.0008
E4 (42)-E4 (123)	0.0094	[0.0076, 0.0112]	0.0105	0.0090
E0 (42)-E0 (123)	−0.0046	[−0.0063, −0.0029]	−0.0046	−0.0119
E6 (42)-E6 (123)	−0.0118	[−0.0151, −0.0085]	−0.0160	−0.0135

S3: full-source ADAST. ADAST was compared with its matched source-only control and, descriptively, with zero-shot E3 and E4 at seed 42.

Table S41: S3: full-source ADAST on the holdout sample (epoch-30 checkpoints). Class-wise differences are pooled over epochs.

Quantity	Difference	95% CI
ADAST-source-only, subject-mean macro-F1	0.0429	[0.0374, 0.0484]
ADAST-source-only, pooled macro-F1	0.0436	[0.0375, 0.0495]
ADAST-source-only, accuracy	−0.0046	[−0.0121, 0.0029]
ADAST-source-only, N3 recall	0.3796	[0.3650, 0.3945]
ADAST-source-only, N1 F1	−0.0190	[−0.0289, −0.0091]
ADAST-source-only, N2 F1	−0.0445	[−0.0535, −0.0359]
ADAST-source-only, REM F1	−0.0568	[−0.0650, −0.0483]
ADAST-E3 (42), subject-mean macro-F1	−0.0222	[−0.0284, −0.0160]
ADAST-E4 (42), subject-mean macro-F1	−0.0361	[−0.0423, −0.0299]

S4: N3 under-detection. The protocol expected an N3→N2 rate above 0.5 for E0 and E3; this expectation was met.

Table S42: S4: N3 recall, precision and N3→N2 rate at seed 42 with subject-cluster bootstrap intervals. The holdout sample contains 61,523 reference N3 epochs.

Arm	N3 recall (95% CI)	N3 precision (95% CI)	N3→N2 rate (95% CI)
E0	0.2565 [0.2359, 0.2772]	0.8811 [0.8284, 0.9248]	0.7230 [0.7014, 0.7440]
E3	0.2549 [0.2347, 0.2749]	0.8814 [0.8283, 0.9254]	0.7268 [0.7066, 0.7474]
E4	0.2742 [0.2539, 0.2942]	0.8831 [0.8331, 0.9250]	0.7114 [0.6910, 0.7321]
E6	0.1987 [0.1788, 0.2190]	0.8468 [0.7889, 0.8959]	0.7713 [0.7500, 0.7924]

S5: present-class macro**-**F1**.** Of the 500 participants, 39 lacked at least one reference class (N1: 16, N3: 34, REM: 21; a participant can lack more than one class). When macro-F1 was averaged only over the classes present in each participant's reference labels, all three confirmatory contrasts remained confirmed (Table S43).

S6: input scale of E6. Within the evaluation windows, the median per-recording standard deviation of the E6 input was 0.992 (interquartile range 0.855–1.000) on the holdout sample, compared with a median of 0.77 on Sleep-EDF. Because E6 standardises each complete recording, the evaluated night of a Sleep-EDF recording, which also contains long daytime segments, reaches the model at a smaller relative scale than an SHHS1 night.

S7: pooled description. Pooling the first and holdout samples (680 participants) gave estimates close to those of each sample. These pooled estimates are descriptive only and were not used for confirmation.

Table S43: S5 and S7: confirmatory contrasts (seed 42) under present-class macro-F1 on the holdout sample and on the pooled 680 participants (descriptive). Δ is the difference in subject-mean macro-F1.

	S5: present-class, n = 500			S7: pooled, n = 680	
Contrast	Δ	95% CI	p_{Holm}	Δ	95% CI
E3−E0	0.0482	[0.0416, 0.0549]	4.3×10^{-37}	0.0452	[0.0398, 0.0506]
E3−E6	0.0296	[0.0231, 0.0359]	3.7×10^{-22}	0.0288	[0.0237, 0.0337]
E3−E2	0.0535	[0.0476, 0.0595]	3.2×10^{-50}	0.0504	[0.0456, 0.0554]

Deviations from the protocol

Two deviations occurred, neither of which changed any prediction. First, inference ran in six parallel CPU worker processes with two threads each instead of one process; the device, deterministic algorithms, checkpoints, ensembling and code path were unchanged, and before the run the new code reproduced the stored fold probabilities of the first sample up to floating-point rounding, with identical predicted stages. Second, the inference launcher was restarted twice while earlier workers were still running. Duplicate workers recomputed some identical units, and four workers stopped with a file-permission error when replacing an output already written by another worker. All 60,000 fold prediction files were present, no temporary file remained, and every file was checked against its recorded input hash, arm, fold and protocol hash when the ensembles were formed.

Supplementary Section S4: approximate input-scale invariance of E3 and E4

This post-hoc analysis was not part of the holdout protocol. It uses only the Sleep-EDF checkpoints and reads no participant data (script `scripts/compare_e3_e4_checkpoints.py`).

Scale dependence of batch normalisation. Let $z = w * x$ be the output of the bias-free first convolution and $\mathrm{BN}_\varepsilon(z) = \gamma(z - \mu_z)/\sqrt{\sigma_z^2 + \varepsilon} + \beta$ the batch-normalisation transform. For a constant $c > 0$, $\mathrm{BN}_\varepsilon(cz) = \mathrm{BN}_{\varepsilon/c^2}(z)$, so with $c = 1/100$ the batch-normalised output of E3 equals that of E4 with ε multiplied by 10^4, for the same convolution weights and batch. In the idealised limit $\varepsilon \to 0$, the layer output does not depend on the scale of z, so $\partial L/\partial(z/100) = 100\,\partial L/\partial z$ and the gradient with respect to w, $(x/100)\,\partial L/\partial(z/100) = x\,\partial L/\partial z$, is the same for both inputs; all later layers receive identical inputs and gradients. Evaluation-mode outputs additionally require the running mean and variance to scale by $1/100$ and $1/10^4$, but both runs initialise the running variance to the same value. With shared seed, parameter initialisation, batch order and AdamW settings, the argument therefore predicts closely similar rather than identical training: finite ε, running-statistic initialisation and finite-precision arithmetic can make the runs diverge, and early stopping can then select different epochs.

Trained weights. Table S44 compares the selected encoder and TCN checkpoints of E3 and E4 for each seed and fold. Cosine similarity is computed between flattened weight tensors with at least two dimensions; the median is taken over the 11 encoder (or all TCN) weight tensors. As a reference, the last column compares seeds 42 and 123 of E3 for the same fold.

Table S44: E3 versus E4 checkpoints from the same seed and fold. Stem: first convolution kernel. Encoder and TCN: median cosine similarity over weight tensors. BN ratio: mean first-layer running variance of E4 divided by that of E3 (expected 10^4 when both select the same epoch). Between seeds: median encoder cosine similarity of E3 seed 42 and seed 123 for the same fold.

Seed	Fold	Epoch E3/E4	Stem	Encoder	TCN	BN ratio	Between seeds
42	0	3/3	0.970	0.848	0.993	9,776	0.002
42	1	10/11	0.931	0.741	0.920	9,502	0.009
42	2	7/7	0.957	0.818	0.988	10,666	0.012
42	3	7/8	0.950	0.756	0.974	11,276	0.004
42	4	8/8	0.948	0.774	0.986	10,427	0.005
42	5	9/7	0.939	0.787	0.954	9,179	0.008
42	6	9/9	0.909	0.808	0.974	10,406	0.003
42	7	10/10	0.938	0.765	0.983	9,489	−0.005
42	8	6/6	0.941	0.833	0.980	10,232	0.008
42	9	7/7	0.954	0.846	0.991	10,250	0.006
123	0	9/9	0.940	0.842	0.987	10,390	0.002
123	1	12/11	0.930	0.739	0.990	8,588	0.009
123	2	14/7	0.938	0.753	0.952	6,215	0.012
123	3	6/6	0.973	0.811	0.942	9,721	0.004
123	4	11/7	0.929	0.696	0.932	7,360	0.005
123	5	11/11	0.943	0.762	0.992	9,208	0.008
123	6	11/6	0.948	0.814	0.938	6,744	0.003
123	7	4/4	0.974	0.863	0.992	9,833	−0.005
123	8	8/8	0.951	0.769	0.965	10,411	0.008
123	9	14/11	0.913	0.699	0.965	6,501	0.006

Across the 20 seed-fold pairs, the stem kernels of E3 and E4 had cosine similarity 0.909–0.974, the median encoder similarity was 0.696–0.863 and the median TCN similarity was 0.920–0.993. Two seeds of the same configuration gave a median encoder similarity of −0.005 to 0.012 and a stem similarity of −0.112 to −0.032. When both runs selected the same epoch, the first-layer running variance of E4 was 9,208–10,666 times that of E3. The running variance of E3 was at least 0.0439, so $\varepsilon = 10^{-5}$ was at most 0.023% of it. Early stopping selected different epochs in 8 of the 20 pairs (3 at seed 42 and 5 at seed 123). For each seed, the trained E3 and E4 models are therefore much closer to each other than to a model trained with another seed, consistent with substantial absorption of the input-scale change; these checkpoint statistics do not separate the contributions of ε, running-statistic initialisation and round-off. Their score difference is compared with the observed seed variability in Section S6.

Supplementary Section S5: post-confirmatory amplitude and AdaBN analyses

Design. The amendment (`docs/SHHS_HOLDOUT_AMENDMENT_A1.md` and `configs/shhs_holdout_amendment_a1.json`) was committed at `ad61aa0` after the confirmatory analysis and before any of its inference; the scripts that package, run and import its GPU computation were committed at `e5a255a`. It evaluates E4 at seed 42 on the 500 holdout participants under five conditions: the band-pass-filtered input multiplied by 0.5, 0.75, 1.5 or 2, with the encoder in evaluation mode and its stored batch-normalisation statistics, and AdaBN. For AdaBN, 40 epochs were drawn without replacement from each recording (random generator seeded with 2041 and the participant identifier; labels not read), and the running mean and variance of each of the ten batch-normalisation layers of the encoder were replaced, in forward order, by the per-channel mean and population variance of that layer's input over the 20,000 sampled epochs, after all earlier layers had been adapted. Affine parameters and the TCN were unchanged. Each fold model was adapted separately and the ten fold probabilities were averaged as in the

parent protocol. The reference condition is the stored E4 seed-42 ensemble of the confirmatory run. Intervals are paired subject-cluster bootstrap intervals (10,000 resamples, seed 2042); no hypothesis tests were specified. The model was chosen after the confirmatory analysis because the band-pass-only form is the recommended form of the pipeline (Supplementary Section S4); no scale factor or adaptation setting depended on holdout results.

Signal amplitude. The per-recording standard deviation of the band-pass-filtered input over the stored epochs had a median of 14.36 μV (interquartile range 11.63–16.82) in the 153 Sleep-EDF recordings and 16.50 μV (13.72–21.26) in the 500 holdout recordings, a ratio of 1.149.

Table S45: Amendment conditions on the holdout sample (n = 500): subject-mean macro-F1 with bootstrap interval and pooled metrics. ADAST and source-only are the full-source epoch-30 arms of the confirmatory run. Intervals use bootstrap seed 2042 and can therefore differ in the last digit from those of the same arms in Table S38 (seed 2040).

Condition	Subject-mean macro-F1 (95% CI)	Pooled macro-F1	Accuracy	κ	W F1	N1 F1	N2 F1	N3 F1	REM F1	N3→N2
E4 (×1)	0.5776 [0.5662, 0.5890]	0.6187	0.7103	0.5896	0.8297	0.3545	0.7471	0.4184	0.7436	71.1%
×0.5	0.4627 [0.4528, 0.4724]	0.4821	0.5950	0.4442	0.8202	0.2558	0.6268	0.0343	0.6733	93.3%
×0.75	0.5256 [0.5149, 0.5362]	0.5600	0.6630	0.5276	0.8272	0.3319	0.6952	0.2294	0.7161	84.2%
×1.5	0.5765 [0.5630, 0.5897]	0.6219	0.7163	0.5903	0.7960	0.3308	0.7730	0.5043	0.7054	60.0%
×2	0.4502 [0.4349, 0.4656]	0.4946	0.6189	0.4464	0.6720	0.3068	0.7187	0.3614	0.4143	48.6%
AdaBN	0.4570 [0.4458, 0.4681]	0.5104	0.6054	0.4497	0.7125	0.2671	0.6823	0.4313	0.4590	54.8%
ADAST	0.5415 [0.5305, 0.5520]	0.5852	0.6798	0.5625	0.8143	0.1208	0.7082	0.7123	0.5701	35.8%
Source-only	0.4986 [0.4899, 0.5071]	0.5416	0.6844	0.5403	0.8153	0.1399	0.7527	0.3731	0.6269	75.3%

Table S46: Differences from unmodified E4 (condition minus E4) with paired subject-cluster bootstrap intervals. Class-wise differences are pooled over epochs.

Condition	Δ subject-mean macro-F1	Δ N3 recall	Δ N3→N2 rate	Δ N1 F1	Δ REM F1
×0.5	−0.1149 [−0.1219, −0.1081]	−0.2565 [−0.2749, −0.2379]	0.2212 [0.1986, 0.2433]	−0.0988 [−0.1090, −0.0880]	−0.0703 [−0.0864, −0.0549]
×0.75	−0.0520 [−0.0559, −0.0482]	−0.1415 [−0.1534, −0.1299]	0.1305 [0.1184, 0.1429]	−0.0226 [−0.0292, −0.0160]	−0.0275 [−0.0350, −0.0202]
×1.5	−0.0011 [−0.0099, 0.0075]	0.0817 [0.0665, 0.0974]	−0.1118 [−0.1284, −0.0962]	−0.0237 [−0.0333, −0.0139]	−0.0382 [−0.0599, −0.0174]
×2	−0.1274 [−0.1412, −0.1137]	−0.0423 [−0.0649, −0.0200]	−0.2253 [−0.2554, −0.1952]	−0.0477 [−0.0614, −0.0340]	−0.3293 [−0.3689, −0.2904]
AdaBN	−0.1206 [−0.1295, −0.1116]	0.0086 [−0.0052, 0.0222]	−0.1638 [−0.1842, −0.1436]	−0.0874 [−0.0997, −0.0754]	−0.2847 [−0.3148, −0.2553]

AdaBN minus ADAST was −0.0844 ([−0.0929, −0.0759]) in subject-mean macro-F1 and −0.3334 ([−0.3487, −0.3185]) in N3 recall; AdaBN minus the source-only control of ADAST was −0.0416 ([−0.0498, −0.0333]) in subject-mean macro-F1. The pre-specified expectation of rising N3 recall and falling N3→N2 rate for factors above 1 was met for the N3→N2 rate, which decreased monotonically over all factors, but not for N3 recall, which peaked at a factor of 1.5. At a factor of 2, 28.2% of reference N3 epochs were classified as wake, compared with 0.9% for unmodified E4.

Deviations from the amendment. First, all 50 condition-by-fold units were computed on a GPU (Tesla T4, PyTorch 2.11.0, deterministic algorithms) instead of the CPU named in the amendment, to shorten computation time. The GPU code reproduces the amendment's prediction and AdaBN functions; before the run it reproduced the stored CPU probabilities of the confirmatory E4 run for two recordings at folds 0 and 9 within 1.6×10^{-6}, inside the locked tolerance of 10^{-5}, with identical predicted stages. Second, a CPU run of the same conditions stopped without an error message after 833 fold files; these files were kept separately and used only to compare devices. Over these 833 files (764,196 epochs), CPU and GPU probabilities differed by at most 1.4×10^{-5}, with identical predicted stages, and the fold-0 AdaBN running variances differed by at most 1.0×10^{-6} in relative terms. Third, the ensemble metadata field for the device retains the script default (CPU), whereas the fold-level metadata record the GPU. None of these deviations changed a condition or the analysis.

Supplementary Section S6: second amendment (A2)

Design. Amendment A2 (`docs/SHHS_HOLDOUT_AMENDMENT_A2.md` and `configs/shhs_holdout_amendment_a2.json`) was written after the confirmatory and A1 analyses and committed at `38eb6dc` before any of its preprocessing, training, inference or analysis. As an independent record, the GitHub server logged an automated test run on this commit at 06:28:06 UTC on 26 October 2026 (GitHub Actions run 40279983451). The tools used on the GPU machine were committed later, at `0954b8c`, without changing the configuration file, whose SHA-256 value is checked before every step and recorded in every A2 output. A2 fixed five descriptive analyses (A-E), with all arms, seeds, the window length, subset sizes, numbers of draws and interpretation rules. Because the holdout labels had already been used in the confirmatory analysis and in A1, A2 is a post-confirmatory, descriptive analysis: it makes no hypothesis tests and does not retest or reinterpret H1–H3. Intervals are paired subject-cluster bootstrap intervals (10,000 resamples, seed 2043).

Training and inference. Fifty new runs were trained on Sleep-EDF with the locked split, runner and recipes: E6R (the E6 recipe on a new input) at seeds 42 and 123, the seeds of the archived E6 runs, and E4 at seeds 2025, 2027 and 2029. The E6R input applies the E3/E6 band-pass filter and clip, followed by a centred rolling z-score over 90 minutes (windows truncated at the recording edges, standard deviation floored at 1 μV); its statistics are computed from the filtered signal without labels. The

window length, about one sleep cycle, gives both cohorts statistics over spans of the same length, but it also removes amplitude changes slower than the window. The Sleep-EDF test folds were opened once, after all 50 runs had finished. The holdout input used the same 500 recordings, channel, resampling, label mapping and evaluation windows as the confirmatory run, and each arm averaged the probabilities of its ten fold models as before. Computation used one Tesla V100-PCIE-16GB, the GPU type of the original campaign, with Python 3.11, PyTorch 2.5.1+cu121 from the project's lock file and deterministic algorithms; three training processes ran in parallel. Holdout inference ran on the GPU after the A2 code had reproduced the stored CPU fold probabilities of E4 seed 42 for two recordings (folds 0 and 9) within 8.3×10^{-7}, inside the locked tolerance of 10^{-5}, with identical predicted stages. No run failed, was resumed or was repeated.

Reproduction check. Before the A2 runs, the archived E4 run for fold 0 at seed 123 was retrained with the A2 runner. The encoder stage was reproduced exactly (18 completed epochs, best epoch 9, validation macro-F1 0.7262). The TCN stage completed the same number of epochs (62) and selected the same epoch (31), but reached a validation macro-F1 of 0.7818 instead of 0.7792; 96.9% of the validation predictions agreed, and logits differed by up to 3.81. As specified, A2 continued and the result is reported as observed. Bitwise reproduction across machines and library builds is therefore not guaranteed for the TCN stage.

A: normalisation scope. Within the evaluation windows, the median per-recording standard deviation of the E6 input was 0.770 (interquartile range 0.672–0.870) on Sleep-EDF and 0.992 (0.855–1.000) on the holdout sample; for E6R it was 0.987 (0.978–0.992) and 0.993 (0.980–0.999). E6R thus presents both cohorts to the model at the same relative scale. Its pooled out-of-fold Sleep-EDF macro-F1 was 0.7702 at seed 42 and 0.7749 at seed 123, against 0.7691 and 0.7780 for E6 and 0.7891 and 0.7887 for E4 (Table S47). The locked interpretation rule required E4−E6R to have an interval above zero at both seeds for the manuscript to retain the statement that preserving calibrated amplitude transferred better than z-scoring; this condition was met (Table S48).

Table S47: A2 arms and the archived arms they are compared with on the holdout sample (n = 500). Intervals use bootstrap seed 2043 and can differ in the last digit from Table S38. Holdout metrics are those of the ten-model ensemble; Sleep-EDF is pooled out-of-fold macro-F1.

Arm	Seed	Subject-mean macro-F1 (95% CI)	Pooled macro-F1	N3 recall	N3→N2	Sleep-EDF macro-F1
E6R	42	0.5412 [0.5311, 0.5512]	0.5734	0.1395	84.2%	0.7702
E6R	123	0.5452 [0.5349, 0.5552]	0.5777	0.1515	82.8%	0.7749
E6	42	0.5344 [0.5228, 0.5459]	0.5711	0.1987	77.1%	0.7691
E6	123	0.5462 [0.5346, 0.5579]	0.5872	0.2018	77.2%	0.7780
E4	42	0.5776 [0.5660, 0.5890]	0.6187	0.2742	71.1%	0.7891
E4	123	0.5682 [0.5565, 0.5796]	0.6082	0.2574	72.6%	0.7887
E4	2025	0.5634 [0.5519, 0.5745]	0.6040	0.2471	73.7%	0.7888
E4	2027	0.5822 [0.5706, 0.5934]	0.6242	0.2658	72.1%	0.7903
E4	2029	0.5686 [0.5574, 0.5795]	0.6102	0.2417	74.4%	0.7854

Table S48: A: normalisation-scope contrasts on the holdout sample at each seed. Pooled and class-wise differences are computed over epochs.

Seed	Contrast	Δ subject-mean macro-F1 (95% CI)	Δ pooled macro-F1 (95% CI)	Δ N3 recall (95% CI)	Δ N3→N2 rate (95% CI)
42	E4−E6R	0.0364 [0.0313, 0.0415]	0.0453 [0.0396, 0.0510]	0.1346 [0.1211, 0.1481]	−0.1305 [−0.1450, −0.1157]
42	E3−E6R	0.0225 [0.0171, 0.0278]	0.0296 [0.0237, 0.0355]	0.1154 [0.1019, 0.1286]	−0.1150 [−0.1295, −0.1003]
42	E6R−E6	0.0068 [0.0012, 0.0124]	0.0023 [−0.0037, 0.0082]	−0.0591 [−0.0741, −0.0447]	0.0706 [0.0552, 0.0864]
123	E4−E6R	0.0230 [0.0177, 0.0282]	0.0305 [0.0248, 0.0363]	0.1059 [0.0931, 0.1187]	−0.1017 [−0.1156, −0.0872]
123	E3−E6R	0.0139 [0.0083, 0.0194]	0.0203 [0.0141, 0.0265]	0.0877 [0.0751, 0.1002]	−0.0864 [−0.1000, −0.0725]
123	E6R−E6	−0.0010 [−0.0071, 0.0051]	−0.0094 [−0.0157, −0.0032]	−0.0503 [−0.0650, −0.0359]	0.0558 [0.0406, 0.0714]

B: five training seeds of E4. Across seeds 42, 123, 2025, 2027 and 2029, holdout subject-mean macro-F1 had a mean of 0.5720, a sample standard deviation of 0.0077 and a range of 0.5634–0.5822, whereas pooled Sleep-EDF macro-F1 had a mean of 0.7884, a standard deviation of 0.0018 and a range of 0.7854–0.7903; the two dispersions use different aggregations (subject-mean ensemble scores versus pooled out-of-fold scores) and are not directly comparable. Nine of the ten pairwise between-seed intervals excluded zero, and the largest absolute difference was 0.0189 (Table S49). Each of the 20 differences between an E4 seed and E0 or E2 at seed 42 or 123 was larger, 0.0313–0.0700 (Table S50; lowest interval bound 0.0263), which meets the locked rule for describing these configuration contrasts as exceeding the observed between-seed range of E4.

Table S49: B: pairwise differences between the five E4 training seeds on the holdout sample (first-named seed minus second).

Contrast	Δ subject-mean macro-F1	95% CI	Δ pooled macro-F1
42–123	0.0094	[0.0076, 0.0112]	0.0105
42–2025	0.0143	[0.0123, 0.0163]	0.0146
42–2027	−0.0046	[−0.0063, −0.0029]	−0.0055
42–2029	0.0090	[0.0069, 0.0111]	0.0084
123–2025	0.0049	[0.0028, 0.0070]	0.0042
123–2027	−0.0140	[−0.0162, −0.0117]	−0.0159
123–2029	−0.0004	[−0.0029, 0.0021]	−0.0020
2025–2027	−0.0189	[−0.0211, −0.0166]	−0.0201
2025–2029	−0.0053	[−0.0072, −0.0034]	−0.0062
2027–2029	0.0136	[0.0116, 0.0155]	0.0139

Table S50: B: E4 at each seed minus E0 and E2 at seeds 42 and 123 (difference in holdout subject-mean macro-F1, with interval).

E4 seed	−E0 (42)	−E0 (123)	−E2 (42)	−E2 (123)
42	0.0606 [0.0541, 0.0671]	0.0560 [0.0495, 0.0628]	0.0654 [0.0598, 0.0711]	0.0455 [0.0405, 0.0508]
123	0.0512 [0.0448, 0.0577]	0.0466 [0.0402, 0.0534]	0.0560 [0.0505, 0.0617]	0.0362 [0.0311, 0.0415]
2025	0.0463 [0.0406, 0.0523]	0.0418 [0.0359, 0.0479]	0.0511 [0.0455, 0.0568]	0.0313 [0.0263, 0.0364]
2027	0.0652 [0.0586, 0.0717]	0.0606 [0.0540, 0.0674]	0.0700 [0.0639, 0.0761]	0.0501 [0.0448, 0.0556]
2029	0.0516 [0.0456, 0.0576]	0.0470 [0.0410, 0.0533]	0.0564 [0.0507, 0.0623]	0.0365 [0.0315, 0.0417]

C: ensembles and single models. Each of the ten fold models of every zero-shot arm was also scored alone on the holdout sample (Table S51). The ensemble exceeded the mean of its single models by 0.0121–0.0327 in subject-mean macro-F1, with every interval above zero, and single models of one arm differed by up to 0.18 (E2, seed 123). N3 recall was slightly higher for single models than for the ensemble in every arm, so ensembling did not cause the N3 deficit. Measured with single models, as on Sleep-EDF, the drop in pooled macro-F1 between the cohorts was 0.2107 and 0.2134 for E3 at seeds 42 and 123 and 0.1916 and 0.2039 for E4, against 0.1875, 0.1902, 0.1704 and 0.1805 with the ensembles.

Table S51: C: single fold models and ten-model ensembles on the holdout sample. Single-model values are the mean (minimum-maximum) over the ten fold models. Gain is the ensemble minus the mean single model in subject-mean macro-F1, with interval.

Arm	Seed	Single subject-mean macro-F1	Ensemble	Gain (95% CI)	N3 recall, single	N3 recall, ensemble
E0	42	0.5049 (0.4574–0.5365)	0.5170	0.0121 [0.0108, 0.0134]	0.2636	0.2565
E0	123	0.5054 (0.4668–0.5556)	0.5216	0.0162 [0.0149, 0.0177]	0.2567	0.2506
E2	42	0.4857 (0.4134–0.5461)	0.5122	0.0265 [0.0239, 0.0291]	0.2568	0.2484
E2	123	0.4997 (0.4077–0.5857)	0.5321	0.0324 [0.0296, 0.0351]	0.2504	0.2402
E3	42	0.5391 (0.4280–0.5911)	0.5637	0.0246 [0.0226, 0.0266]	0.2637	0.2549
E3	123	0.5356 (0.4974–0.5708)	0.5591	0.0235 [0.0214, 0.0255]	0.2499	0.2392
E4	42	0.5567 (0.5111–0.5951)	0.5776	0.0209 [0.0191, 0.0228]	0.2809	0.2742
E4	123	0.5442 (0.4763–0.5710)	0.5682	0.0240 [0.0221, 0.0258]	0.2640	0.2574
E4	2025	0.5393 (0.4882–0.5605)	0.5634	0.0241 [0.0222, 0.0259]	0.2560	0.2471
E4	2027	0.5539 (0.5195–0.5855)	0.5822	0.0284 [0.0262, 0.0305]	0.2716	0.2658
E4	2029	0.5465 (0.4770–0.5727)	0.5686	0.0221 [0.0200, 0.0242]	0.2484	0.2417
E6	42	0.5017 (0.4463–0.5452)	0.5344	0.0327 [0.0303, 0.0351]	0.2126	0.1987
E6	123	0.5148 (0.4481–0.5462)	0.5462	0.0314 [0.0290, 0.0338]	0.2117	0.2018
E6R	42	0.5199 (0.4861–0.5466)	0.5412	0.0213 [0.0196, 0.0230]	0.1538	0.1395
E6R	123	0.5293 (0.5069–0.5644)	0.5452	0.0159 [0.0145, 0.0174]	0.1631	0.1515

D: sample size of the EM prior estimate. The ten seed-123 E3 models were applied to the holdout sample with the archived per-fold temperatures and source priors; only the data used to estimate the target priors changed, and the ten adjusted fold probabilities were averaged before argmax. EM converged for every fold in every condition. The reference share of N3 among holdout epochs was 0.134. The archived priors from the five adaptation recordings placed N3 at the floor of 10^{-12} in every fold, and the adjusted ensemble predicted no N3. With priors from the 20,000 epochs of the AdaBN sample or from all 500 recordings, the fold-mean N3 prior rose only to 0.030 and 0.029, about a fifth of the reference share, and N3 recall fell from 0.240 for the temperature-scaled ensemble without adjustment to 0.134 and 0.129 (Table S52). Random subsets converged to the same value as they grew: the median N3 prior was 0.028, 0.029 and 0.029 for 5, 20 and 100 recordings, and its 95% range narrowed from 0–0.109 to 0.020–0.042. Larger samples therefore removed the variability of the estimate but not its bias. Both

large-sample priors lay below half the reference share and lowered N3 recall, which meets the locked rule attributing the EM collapse to the failure of the label-shift assumption under feature shift rather than to the five-recording sample.

Table S52: D: EM prior correction for the seed-123 E3 ensemble on the holdout sample (n = 500; reference N3 share 0.134). N3 prior is the mean over the ten folds. Random subsets give the median and the 2.5–97.5th percentiles over 200 draws. Per-record: each recording estimates its own priors; its N3 prior is the median over recordings.

Prior estimated from	N3 prior	Predicted N3 share	N3 recall	Pooled macro-F1	Subject-mean macro-F1
No adjustment (raw ensemble)	n/a	0.036	0.2392	0.5980	0.5591
No adjustment (temperature only)	n/a	0.037	0.2403	0.5985	0.5594
Five adaptation recordings (archived)	$< 10^{-11}$	0.000	0.0000	0.5086	0.4914
AdaBN sample (20,000 epochs)	0.0302	0.021	0.1342	0.5495	0.5173
All 500 recordings	0.0289	0.020	0.1286	0.5479	0.5162
Each recording separately	0.0078	0.035	0.2185	0.5133	0.4477
5 random recordings	0.028 [0.000, 0.109]	0.019	0.120 [0.000, 0.302]	0.531 [0.461, 0.590]	0.502 [0.442, 0.552]
20 random recordings	0.029 [0.004, 0.058]	0.020	0.129 [0.010, 0.208]	0.543 [0.502, 0.575]	0.512 [0.482, 0.539]
100 random recordings	0.029 [0.020, 0.042]	0.020	0.129 [0.094, 0.169]	0.548 [0.535, 0.561]	0.516 [0.506, 0.526]

E: training cost. Completed epochs and optimiser steps were read from the final training checkpoints of every stage of every fold in the archived seed-123 campaign, and wall-clock times from its monitoring records. Time per optimiser step was measured on the A2 GPU through the training loop itself, with the optimiser, loss, batch size and gradient clipping of each stage, using prebuilt batches (sequence batches with the record lengths of the fold-0 training records): 20 warm-up steps and three rounds of 200 timed steps. Mean step times were 6.67 ms for one baseline CNN branch (range over rounds 6.31–6.94), 16.09 ms for ResNet-1D (15.58–16.49), 83.3 ms for the BiLSTM (81.4–85.0) and 30.9 ms for the TCN (30.6–31.2). No baseline CNN approached its limit of 1,000 epochs: the 150 CNNs stopped after 29.1 epochs on average, and the ResNet-1D encoders of E3 after 19.0 of at most 40. Because the baseline trains 15 encoder networks per fold, it ran 22.5 times as many optimiser steps as E3, but each of its steps took 0.41 times as long on average (Table S53). The time not accounted for by optimiser steps (validation, feature extraction, data loading and checkpointing) was 40.5% of the wall-clock time for E0 and 33.9% for E3.

Table S53: E: decomposition of ten-fold training time (seed 123, Tesla V100). Steps: optimiser steps summed over all stages and folds. Step time: the steps multiplied by the measured mean time per step. Remainder: wall-clock time not accounted for by step time. Ratios are E0 divided by the configuration.

ID	Networks per fold	Epochs	Steps	Step time (h)	Wall clock (h)	Remainder	Steps ratio	Wall-clock ratio
E0	16	4,635	10,692,557	19.99	33.59	40.5%	1.0	1.0
E2	2	735	401,776	1.83	2.75	33.3%	26.6	12.2
E3	2	857	474,809	2.17	3.28	33.9%	22.5	10.2
E4	2	790	425,612	1.94	2.93	33.7%	25.1	11.5
E6	2	753	344,928	1.58	2.41	34.4%	31.0	13.9

Deviations from the amendment. Analysis D was run on a local CPU after, rather than before, the GPU stage; it reads only archived E3 seed-123 probabilities and calibration fits and none of the A2 outputs, so the order did not affect it. No other deviation occurred.

Supplementary Section S7: power spectra of the model inputs

This post-hoc description was not part of the holdout protocol or of amendments A1 and A2. It reads the stored evaluation-window epochs of the unfiltered and band-pass inputs for the 153 Sleep-EDF recordings and the 500 holdout SHHS1 recordings, loads no model and computes no performance metric; stage labels enter only through the evaluation window and the exclusion of Movement and Unknown epochs (script `scripts/analyze_cohort_psd.py`). For each recording, the power spectral density of every valid 30 s epoch was estimated at 100 Hz by Welch's method (20 s Hann segments, 50% overlap, segment mean removed) and averaged over epochs. Power fractions are sums over the frequency bins below 0.5 Hz, from 0.5 to below 30 Hz and from 30 to 50 Hz, divided by the total without the zero-frequency bin (Table S54).

Table S54: Share of signal power in three frequency ranges, as median and interquartile range over recordings (%). The band-pass input retains a small share below 0.5 Hz because the filter attenuation at its corner frequency is finite.

Cohort	Input	Below 0.5 Hz	0.5 to 30 Hz	Above 30 Hz
Sleep-EDF (n = 153)	Unfiltered	26.4 [21.4, 33.2]	68.4 [60.0, 73.4]	3.1 [1.4, 6.9]
	Band-pass	1.3 [1.0, 1.6]	98.6 [98.3, 98.9]	0.08 [0.04, 0.13]
SHHS1 holdout (n = 500)	Unfiltered	21.0 [14.4, 28.9]	77.7 [69.9, 84.0]	1.1 [0.7, 1.8]
	Band-pass	0.5 [0.4, 0.6]	99.4 [99.3, 99.5]	0.07 [0.05, 0.11]

Between 0.1 and 0.2 Hz, the median SHHS1 spectrum of the unfiltered input exceeded that of Sleep-EDF by 3.2–8.8 dB, and from 40 to 50 Hz it was 7.7–19.6 dB lower; within 0.5–30 Hz the differences ranged from −2.5 to +4.4 dB for the unfiltered input and from −2.3 to +4.4 dB for the band-pass input. The 0.05 Hz bin adjoins the removed segment mean and is not interpreted. The decline of the unfiltered SHHS1 spectrum above 40 Hz coincides with the transition band of the anti-aliasing filter applied when the 125 Hz recordings were resampled to 100 Hz. The spectra are shown in Fig. 4 of the main text.

Supplementary Section S8: stage-wise amplitude and slow-wave power of the band-pass input

This post-hoc description was not part of the holdout protocol or of amendments A1 and A2. It reads the stored evaluation-window epochs of the band-pass input and their reference stages for the 153 Sleep-EDF recordings and the 500 holdout SHHS1 recordings, and uses no model and no prediction (script `scripts/analyze_stage_spectra.py`). For every valid 30 s epoch it computes the standard deviation in microvolts and the power between 0.5 and 4 Hz (Welch estimate at 100 Hz with 4 s Hann segments and 50% overlap, summed over the band and expressed in dB relative to 1 μV^2). A recording enters the summary of a stage if it has at least ten epochs of that stage (Table S55).

Table S55: Epoch standard deviation and 0.5–4 Hz power of the band-pass input by reference stage: median and interquartile range over recordings of the per-recording median over epochs. n is the number of recordings with at least ten epochs of the stage.

Cohort	Stage	n	Standard deviation (μV)	0.5–4 Hz power (dB)
Sleep-EDF	W	153	15.17 [11.80, 17.89]	22.12 [19.68, 24.02]
	N1	153	8.14 [6.66, 9.99]	16.20 [14.63, 18.38]
	N2	153	11.46 [9.39, 13.74]	19.95 [18.45, 21.63]
	N3	117	20.84 [16.31, 23.33]	25.81 [23.92, 27.01]
	REM	153	7.40 [6.31, 8.95]	15.59 [14.34, 17.06]
SHHS1 holdout	W	500	12.97 [10.62, 16.30]	17.95 [15.80, 20.54]
	N1	453	9.17 [7.97, 10.71]	15.63 [14.43, 17.08]
	N2	500	13.00 [11.71, 14.75]	20.25 [19.35, 21.37]
	N3	455	18.83 [16.94, 21.11]	24.65 [23.75, 25.59]
	REM	478	8.57 [7.56, 9.89]	15.80 [14.62, 16.97]

Within the recordings that had at least ten epochs of both stages, the ratio of the N3 to the N2 median standard deviation was 1.62 [1.44, 1.80] on Sleep-EDF (n = 117) and 1.43 [1.34, 1.54] on the holdout sample (n = 455), and the N3 minus N2 difference in 0.5–4 Hz power was 4.90 [3.67, 5.97] dB and 4.22 [3.57, 4.90] dB.

Table S56 describes how well each quantity alone separates reference N3 from reference N2 epochs, pooled over the epochs of a cohort (69,132 N2 and 13,039 N3 epochs in Sleep-EDF; 207,848 and 61,523 in the holdout sample). The rule assigns N3 to an epoch at or above a threshold; the threshold of a cohort is the value, among 999 quantiles of its pooled N2 and N3 epochs, that maximises the sum of the two rates. The analysis considers these two stages only and is not a five-stage classifier.

Table S56: Separation of reference N3 from reference N2 epochs by a single quantity of the band-pass input. AUC is the area under the receiver operating characteristic curve within the cohort. The last two columns apply the stated threshold to the epochs of the evaluated cohort.

Quantity	Evaluated cohort	Threshold from	AUC	N3 at or above	N2 below
Standard deviation	Sleep-EDF	Sleep-EDF (16.51 μV)	0.940	85.6%	87.0%
	SHHS1 holdout	SHHS1 holdout (15.71 μV)	0.892	90.3%	77.2%
	SHHS1 holdout	Sleep-EDF (16.51 μV)		84.0%	82.2%
	Sleep-EDF	SHHS1 holdout (15.71 μV)		88.8%	83.3%
0.5–4 Hz power	Sleep-EDF	Sleep-EDF (23.62 dB)	0.950	87.2%	88.0%
	SHHS1 holdout	SHHS1 holdout (22.71 dB)	0.913	90.3%	81.6%
	SHHS1 holdout	Sleep-EDF (23.62 dB)		77.4%	88.8%
	Sleep-EDF	SHHS1 holdout (22.71 dB)		93.1%	80.9%

Supplementary Section S9: whole-night inference time on a CPU

This post-hoc timing was not part of the holdout protocol or of amendments A1 and A2 (script `scripts/benchmark_cpu_whole_night.py`). It calls the inference function of the holdout evaluation, which encodes the epochs in batches of 256, applies the sequence model to the whole night and takes the softmax, for the locked fold-0, seed-42 model of E0 and of E3 on the holdout recording of median length (932 epochs, 7.8 h). E2, E4, E6 and E6R share the architecture of E3. Disk I/O, preprocessing and model loading are excluded. The processor was an Intel Core i5-13500H under Windows 11 with Python 3.11.9 and PyTorch 2.5.1 (CPU build). For each thread setting, one untimed warm-up run per model was followed by five timed runs per model in alternating order (Table S57). The times depend on the machine and describe one fold model; a ten-model ensemble takes ten times as long.

Table S57: Whole-night CPU inference of one fold model on a 932-epoch recording: median and range of five runs.

Threads	Configuration	Time (s)	Epochs s^{-1}	E0 time / E3 time
1	E0	4.31 [4.27, 4.34]	216	
1	E3	1.38 [1.35, 1.39]	676	3.13
4	E0	1.75 [1.74, 1.86]	534	
4	E3	0.55 [0.54, 0.57]	1,699	3.18

Provenance

The historical SHHS protocol matches the original run-manifest hash (‘165d7cdf...fe93‘), not the expanded post-run audit (‘9541e233...fe9‘). Table S21 reuses seed-123 diagnostics matched to the EDF campaign and 20 E0/E3 fold-prediction files, and the SHHS1 extension manifest and 360 E0/E3 ensemble files. The retained files are linked to these analyses by their SHA-256 hashes. SHHS data and predictions remain externally held; E6 uses retained predictions. Snapshot reconciliation is documented in `Reports/SHHS_PROTOCOL_PROVENANCE.md`.